A Project Report on

DEEP FRUITVEG IDENTIFICATION AND CLASSIFICATION

Industrial Internship Project report submitted in partial fulfilment of the Requirements for the award of the degree in

BACHELOR OF TECHNOLOGY

IN

CSE (ARTIFICIAL INTELLIGENCE & DATA SCIENCE)

BY

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NH-5, CHOWDAVARAM, GUNTUR - 522019

2022 -2026

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This is to certify that the Industrial Internship Project work entitled **DEEP FRUITVEG IDENTIFICATION AND CLASSIFICATION** being submitted by **G. NAGENDRA BABU (228X1A4509), B.VENKATA KOTI REDDY (228X1A4550), A. MANOJ KUMAR REDDY(228X1A4548), U. MUNI SEKHAR (228X1A64)** in partial fulfilment for the award of the Degree of Bachelor of Technology in CSE (Artificial Intelligence & Data Science) in the Kallam Haranadhareddy Institute of Technology is a record of bona fide work carried out by them.

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DECLARATION

We G. NAGENDRA BABU (228X1A4509), B. VENKATA KOTI REDDY (228X1A4550), A. MANOJ KUMAR REDDY (228X1A4548), U. MUNI SEKHAR (228X1A4564), hereby declare that the project report titled "**DEEP FRUITVEG IDENTIFICATION AND CLASSIFICATION**" under the guidance of **Mr. R. ANANDA BABU** is submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in CSE (Artificial Intelligence & Data Science).

This is a record of bona fide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project have not been submitted to any other university for the award of any degree.

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ACKNOWLEDGEMENT

We profoundly grateful to express our deep sense of gratitude and respect towards **Sri. KALLAM MOHAN REDDY**, **Chairman, KHIT**, **Guntur** for his precious support in the college**.**

We are thankful to **Dr. M. UMA SANKAR REDDY**, Director, KHIT, GUNTUR for his encouragement and support for the completion of the project.

We are much thankful to **Dr. B. SIVA BASIVI REDDY**, Principal, KHIT, GUNTUR for his support during and until the completion of the project.

We are greatly indebted to **Dr. A. HANUMAT PRASAD, M.Tech, PhD** Professor & Head of the department, CSE (Artificial Intelligence & Data Science), KHIT, GUNTUR for providing the laboratory facilities fully as and when required and for giving us the opportunity to carry the project work in the college during Industry Internship

We extend our deep sense of gratitude to our Internal Guide **Mr. R. ANANDA BABU**, who helped us in each step of our project.

We are also thankful to our Project Coordinator **Mr. R. ANANDA BABU,** who helped Us in each step of our Project.

We are also thankful to our other Faculty Members & Support Staff for their valuable suggestions, guidelines and constructive ideas in each and every step, which was indeed of great help towards the successful completion of our project.

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ABSTRACT

The classification of fruits and vegetables plays a vital role in agriculture, the food industry, and retail management. Traditional methods of sorting and identifying fruits and vegetables are manual, time-consuming, and often prone to human error, leading to inefficiencies and inconsistencies in quality control. To overcome these challenges, this project introduces a **Deep Learning-based Fruits and Vegetables Classification System** that employs **Convolutional Neural Networks (CNNs)** to automatically recognize and categorize different types of fruits and vegetables from images.

The proposed system focuses on analyzing key visual features such as color, texture, and shape to ensure accurate classification, even under varying lighting conditions and complex backgrounds. A diverse dataset containing multiple fruit and vegetable categories is collected, preprocessed, and augmented to enhance the model's robustness and generalization capability. The model is developed and trained using **TensorFlow and Keras frameworks**, and its performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

This deep learning approach significantly improves the speed, reliability, and accuracy of classification while reducing the need for manual intervention. It provides an efficient solution for automating sorting, quality inspection, and inventory management processes in agriculture and retail industries. The successful implementation of this system demonstrates the potential of artificial intelligence in advancing smart farming and intelligent food supply chain operations.

Overall, this project highlights the potential of deep learning in transforming traditional agricultural and retail practices through automation and intelligent decision-making. By reducing human dependency, increasing efficiency, and improving consistency, the **Deep Fruits and Vegetables Classification System** provides a practical and scalable solution for modern smart farming, food processing, and inventory management applications. This work lays the foundation for further research and development in agricultural automation, paving the way toward smarter, more efficient, and technology-driven food production systems.

INTRODUCTION

INTRODUCTION

In the modern era of technology and automation, artificial intelligence (AI) and machine learning (ML) have revolutionized various sectors, including healthcare, finance, transportation, and agriculture. Among these, agriculture plays a crucial role in sustaining human life, and the integration of AI technologies has opened new opportunities for improving productivity, quality control, and efficiency. One of the critical areas in agricultural automation is the **classification and recognition of fruits and vegetables**, which is vital for tasks such as sorting, grading, packaging, and inventory management. Traditionally, these tasks have been performed manually by human workers, which often results in errors, inconsistency, and time consumption, especially when dealing with large-scale production.

The need for an **automated, accurate, and reliable system** for identifying fruits and vegetables has led to the adoption of **deep learning** and **computer vision** techniques. Deep learning, particularly **Convolutional Neural Networks (CNNs)**, has shown remarkable success in the field of image classification due to its ability to learn complex visual patterns automatically. By training CNN models on large datasets of fruit and vegetable images, it becomes possible to classify different categories based on features such as colour, texture, shape, and size, without requiring manual feature extraction. This makes deep learning-based systems far more efficient and scalable than traditional image processing approaches.

The **Deep Fruits and Vegetables Classification System** developed in this project aims to provide a highly accurate and automated solution for classifying various fruit and vegetable species using image data. The project focuses on building a comprehensive dataset consisting of different fruits and vegetables captured under diverse environmental conditions, including changes in lighting, background, and angle. This dataset is pre-processed and augmented to enhance model robustness and generalization. The system is trained and evaluated using deep learning frameworks such as **TensorFlow** and **Keras**, ensuring a modern and efficient implementation.

The project also explores the importance of transfer learning by utilizing pre-trained CNN models such as **VGG16**, **ResNet50**, or **InceptionV3**, which help achieve higher accuracy with limited computational resources and smaller datasets. Once trained, the model can be integrated into real-world applications like automated sorting machines, mobile apps for fruit identification, or retail systems for checkout and quality verification.

In conclusion, the **Deep Fruits and Vegetables Classification System** represents a significant step forward in agricultural automation and intelligent food management. It combines the power of computer vision and deep learning to solve real-world problems efficiently. The project not only enhances operational efficiency and accuracy but also reduces dependency on manual labor. Through this system, agriculture can move toward a more advanced, data-driven, and technology-supported framework that ensures productivity, quality, and sustainability. This project lays the foundation for future innovations in **AI-based agricultural solutions**, contributing to the broader vision of **smart farming and intelligent automation**.

Project Idea

The idea of the **Deep Fruits and Vegetables Classification Project** is to develop an intelligent system that can automatically identify and classify different types of fruits and vegetables using **deep learning and image processing techniques**. The project aims to replace manual sorting and recognition methods, which are often slow, inconsistent, and prone to human error, with an automated, accurate, and efficient solution.

The core concept involves capturing or collecting images of various fruits and vegetables, preprocessing them, and then training a **Convolutional Neural Network (CNN)** model to recognize distinct visual features such as colour, texture, and shape. The trained model can then classify each image into its respective category, such as apple, banana, tomato, or carrot. The system is built using **TensorFlow and Keras frameworks**, which provide efficient tools for model training, testing, and optimization.

This project can be extended into practical applications such as **automated sorting systems**, **smart retail billing**, **quality inspection**, and **mobile applications** for farmers and vendors. By implementing deep learning in this domain, the project demonstrates how artificial intelligence can transform traditional agricultural and retail processes into **smart, automated, and data-driven systems**.

Once the model is trained, it is evaluated based on metrics such as **accuracy, precision, recall, and F1-score** to determine its reliability and efficiency. The final system will be able to classify unseen images of fruits and vegetables accurately and consistently. The trained model can then be integrated into real-world applications such as **automated sorting machines in food industries**, **mobile applications for farmers**, **retail billing systems**, or **smart inventory management systems**.

1.2. Motivation of the Project

The motivation behind developing the **Deep Fruits and Vegetables Classification System** comes from the growing need to automate agricultural and retail processes that are traditionally done manually. Manual sorting and classification of fruits and vegetables are time-consuming, inconsistent, and prone to human error. With the increasing demand for efficiency, accuracy, and large-scale production, there is a strong need for intelligent systems that can perform these tasks automatically.

Advancements in **Artificial Intelligence (AI)** and **Deep Learning**, especially **Convolutional Neural Networks (CNNs)**, provide powerful tools for image recognition and classification. These technologies can analyse features such as colour, texture, and shape to accurately classify different fruits and vegetables. Implementing such a system reduces human labour, increases processing speed, ensures quality control, and minimizes post-harvest losses.

Overall, the project is motivated by the vision of achieving **automation, accuracy, and efficiency** in agriculture through modern AI techniques, contributing to the development of **smart farming and intelligent food processing systems**.

1.3. Problem Statement

In today's agricultural and food processing industries, the accurate classification and identification of fruits and vegetables play a crucial role in ensuring quality control, efficient packaging, and effective inventory management. Traditionally, these tasks are performed manually by human workers who sort and classify produce based on their physical characteristics such as colour, size, shape, and texture. However, manual sorting is **time-consuming, labour-intensive, inconsistent, and prone to human error**, especially when handling large quantities of produce. The subjectivity involved in human judgment often leads to misclassification, affecting product quality, pricing, and overall productivity.

Furthermore, the increasing demand for **automation and efficiency** in agricultural operations has made it necessary to adopt intelligent systems that can perform classification tasks quickly and accurately. Existing traditional image processing methods rely on predefined rules and handcrafted features, which are not effective in handling variations in lighting conditions, orientation, occlusion, and complex backgrounds. Fruits and vegetables often exhibit **high intra-class variability** (differences within the same type, such as color variations among apples) and **low inter-class differences** (similar appearance between different items, such as tomatoes and red apples). These visual similarities make the task of classification even more challenging for conventional techniques.

To overcome these limitations, there is a need for a **deep learning-based automated classification system** that can accurately recognize and categorize fruits and vegetables from digital images. Deep learning models, particularly **Convolutional Neural Networks (CNNs)**, have demonstrated exceptional performance in image recognition tasks by automatically learning important features from raw images without manual feature extraction.

Therefore, the problem addressed in this project is the **lack of an efficient, accurate, and automated system** for fruit and vegetable classification that can operate under varying conditions and deliver consistent results. The goal is to design and develop a deep learning-based model that can classify different fruits and vegetables based on their visual features, using datasets collected from real-world environments. The system aims to improve the speed, precision, and scalability of classification processes, thereby reducing manual effort and enhancing productivity in agriculture, food industries, and retail sectors.

1.4. Statement of scope

The **Deep Fruits and Vegetables Classification Project** focuses on developing a deep learning-based system that can automatically identify and classify various fruits and vegetables from digital images with high accuracy. The project covers key stages such as **data collection, preprocessing, model training, testing, and performance evaluation** using frameworks like **TensorFlow** and **Keras**.

The system aims to assist in **automating sorting, grading, and quality inspection processes** in agricultural and retail sectors. It is designed to handle image variations such as lighting, background, and orientation to ensure robust performance. However, the project does not cover aspects like disease detection, ripeness analysis, or integration with industrial hardware systems.

In summary, the project's scope is limited to developing a software-based image classification model capable of accurate fruit and vegetable recognition, which can later be integrated into real-world applications for smart agriculture and retail automation.

Goals and Objectives

The main goal of the **Deep Fruits and Vegetables Classification Project** is to design and develop an intelligent, automated system capable of accurately classifying different types of fruits and vegetables using **deep learning techniques**. The project aims to reduce human intervention, improve accuracy, and enhance efficiency in agricultural and retail operations through the use of **Convolutional Neural Networks (CNNs)** and modern image processing methods.

Goals:

1. To automate the process of fruit and vegetable classification using deep learning and computer vision.
2. To enhance the efficiency, speed, and reliability of sorting and grading in agricultural and food industries.

3. To develop a scalable and adaptable model that can be integrated into real-world systems such as sorting machines, mobile apps, or retail systems.
4. To promote the adoption of AI-based smart farming and precision agriculture practices.

Objectives:

1. To collect and prepare a large, diverse dataset of fruit and vegetable images for model training and testing.
2. To preprocess and augment the dataset to improve image quality and model generalization.
3. To design, train, and optimize a **Convolutional Neural Network (CNN)** model using frameworks like **TensorFlow** and **Keras**.
4. To evaluate the model's performance using metrics such as accuracy, precision, recall, and F1-score.
5. To develop a user-friendly prototype that demonstrates the real-time classification of fruits and vegetables.

STUDY PHASE

LITERATURE SURVEY

2.1 Literature Survey

The classification of fruits and vegetables using artificial intelligence and computer vision has gained significant attention in recent years due to the increasing demand for automation in agriculture, food processing, and retail sectors. Various studies have explored different approaches to fruit and vegetable recognition, primarily focusing on image processing techniques, machine learning algorithms, and deep learning architectures.

Early approaches relied on **traditional image processing techniques**, such as colour-based segmentation, shape analysis, and texture recognition, to classify fruits and vegetables. For example, methods using histogram analysis, edge detection, and colour thresholding were applied to distinguish between fruit types based on their visual features. However, these approaches often required manual feature extraction and struggled to handle variations in lighting, occlusion, or background complexity, limiting their practical applicability.

Previous Research Work:

Recent studies have significantly advanced the application of deep learning techniques in the classification of fruits and vegetables. These efforts have focused on improving accuracy, scalability, and real-time performance across various domains, including agriculture, retail, and food quality assessment.

1. YOLOv4-Based Freshness Classification
2. MobileNetV2 for Lightweight Classification
3. Dual-Model System for Classification and Ripeness Assessment
4. Multi-Input CNN for Defect Detection
5. DenseNet for Quality Grading

Existing System

Earlier technological solutions have attempted to automate classification using **image processing and machine learning techniques**. Systems based on **colour histogram analysis, shape detection, and texture analysis** were developed to distinguish fruits and vegetables. These methods typically require **manual feature extraction** and predefined rules for classification. While they showed moderate success in controlled environments, they often fail in real-world scenarios where lighting conditions, backgrounds, occlusion, and fruit variations are unpredictable.

Disadvantages of Existing System

Despite the advancements in automated fruit and vegetable classification, the existing systems have several limitations that restrict their effectiveness and real-world applicability. Some of the major disadvantages include:

1. **Manual Dependency:**
   Traditional methods rely heavily on human workers for sorting and classification, which is time-consuming, labour-intensive, and prone to human error.

2. **Inconsistent Accuracy:**
   Manual inspection and rule-based image processing techniques often produce inconsistent results due to subjective human judgment or varying environmental conditions such as lighting, background, and occlusion.

3. **Limited Scalability:**
   Existing systems, particularly those based on traditional image processing or machine learning with handcrafted features, struggle to handle large-scale datasets or multiple fruit and vegetable classes efficiently.

4. **High Complexity for Feature Extraction:**
   Conventional machine learning approaches depend on manually engineered features (colour, shape, texture), which require domain expertise and may not generalize well across different datasets.

5. **Poor Performance in Real-World Conditions:**
   Variations in illumination, camera angles, overlapping fruits, and environmental backgrounds often reduce the performance of existing systems, making them unsuitable for practical deployment.

6. **Computational Requirements:**
   While deep learning-based systems provide higher accuracy, some pre-trained models require high computational power, which may limit their deployment in resource-constrained environments like mobile devices or small-scale farms.

2.2 Feasibility Study

A feasibility study is essential to determine whether the proposed **Deep Fruits and Vegetables Classification System** can be successfully developed, implemented, and sustained within technical, economic, and operational constraints. The feasibility of this project is assessed in the following dimensions:

1.Technical Feasibility

The technical feasibility evaluates whether the project can be developed using the available technologies and resources:

- **Deep Learning Frameworks:** Tools like **TensorFlow** and **Keras** provide robust platforms for designing, training, and deploying Convolutional Neural Networks (CNNs).
- **Hardware Requirements:** Moderate GPU-equipped systems or cloud computing platforms can handle training and testing of models efficiently.

- **Integration Possibility:** The system can be integrated into mobile apps, retail billing software, or automated sorting machinery using APIs and user-friendly interfaces.

2. Economic Feasibility

Economic feasibility assesses whether the project can be developed and maintained within a reasonable budget:

- **Software Costs:** Open-source frameworks like TensorFlow and Keras reduce software expenses.
- **Hardware Costs:** Standard GPUs or cloud-based computing resources provide affordable options for training models.

3. Operational Feasibility

Operational feasibility evaluates whether the system can be effectively deployed and used by intended users:

- **Ease of Use:** The system can provide user-friendly interfaces for farmers, retail staff, or warehouse operators.
- **Scalability:** Can be scaled to include additional fruit and vegetable classes or extended to ripeness and quality assessment.

4. Schedule Feasibility

The development and deployment of the system are achievable within a reasonable timeframe:

- Dataset collection and preprocessing can be completed in the initial phase.
- Model development, training, and testing can be conducted in parallel using GPU resources.

Overall Feasibility

Considering the technical, economic, operational, and schedule aspects, the proposed **Deep Fruits and Vegetables Classification System** is **highly feasible**. Modern deep learning tools, accessible datasets, affordable computing resources, and clear user benefits make this project practical and implementable in real-world agricultural and retail applications.

PROPOSED SYSTEM

PROPOSED SYSTEM

3.1 Proposed System

The proposed system aims to develop an automated deep learning-based solution for the accurate classification of fruits and vegetables. Unlike existing systems that rely on manual inspection or traditional image processing methods, this system leverages Convolutional Neural Networks (CNNs) to automatically learn and recognize visual features such as color, texture, shape, and size. The system is designed to be robust, scalable, and capable of handling real-world challenges such as varying lighting conditions, complex backgrounds, and overlapping produce.

Key Features of the Proposed System:

1. Automation: The system eliminates manual intervention, providing faster and more consistent classification.
2. High Accuracy: CNN-based deep learning models allow for precise recognition of multiple fruit and vegetable classes.
3. Scalability: The system can be extended to include additional classes, ripeness detection, or defect identification.
4. Real-Time Operation: With optimization and deployment, the system can classify images in real-time, making it suitable for retail, agricultural, and warehouse applications.
5. User-Friendly Interface: The system can be integrated into mobile applications or desktop platforms for easy accessibility.

System Workflow:

The proposed system follows a structured workflow to ensure accuracy and efficiency:

1. Data Collection: Images of various fruits and vegetables are collected from open-source datasets or captured manually under different lighting and background conditions.

2. Data Preprocessing: The images are resized, normalized, and augmented (rotation, flipping, brightness adjustments) to enhance model robustness and prevent overfitting.
3. Model Training: A CNN model is designed and trained using frameworks like TensorFlow and Keras, learning hierarchical features directly from images.
4. Validation and Testing: The trained model is validated and tested using unseen data, with performance metrics such as accuracy, precision, recall, and F1-score.
5. Deployment: The model can be deployed on desktop applications, mobile apps, or integrated into automated sorting systems for real-time classification.
6. Output: The system identifies the type of fruit or vegetable in the input image and provides the classification result instantly.

3.2.Methodology

The methodology defines the systematic approach followed to develop the Deep Fruits and Vegetables Classification System. It outlines the steps involved, from data collection to model deployment, ensuring the project is implemented efficiently and effectively. The methodology is divided into several key phases:

1. Data Collection

The first step involves gathering a diverse and high-quality dataset of fruit and vegetable images. The images can be collected from:

- Publicly available datasets from research repositories.
- Online sources to include variety in shape, color, and size.

A diverse dataset ensures that the model can generalize well and handle real-world variability.

2. Data Preprocessing

Raw images collected from various sources need preprocessing to improve model performance:

- Resizing: Standardize all images to a fixed size for uniformity in model input.
- Normalization: Scale pixel values to a range suitable for deep learning models.
- Data Augmentation: Techniques such as rotation, flipping, zooming, brightness adjustment, and cropping are applied to increase dataset diversity and prevent overfitting.
- Labeling: Each image is labeled correctly with its corresponding fruit or vegetable class.

3. Model Training

The CNN model is trained on the pre-processed dataset using TensorFlow and Keras:

- Loss Function: Categorical cross-entropy for multi-class classification.
- Optimizer: Adam optimizer to adjust weights and minimize loss.
- Training Process: The model learns from the training dataset while monitoring performance on the validation set to avoid overfitting.

4. Model Evaluation

After training, the model is tested using a separate test dataset to evaluate its performance using metrics such as:

- Accuracy: Overall correctness of classification.
- Precision and Recall: For evaluating class-specific performance.
- F1-Score: Harmonic mean of precision and recall for balanced evaluation.

5. Deployment

Once the model achieves satisfactory performance, it can be deployed for real-time applications:

- Integration into desktop or web applications for fruit and vegetable recognition.
- Deployment in mobile applications for farmers and retailers.

- Integration with automated sorting machines for large-scale operations.

7. Maintenance and Updates

The system can be continuously improved by:

- Updating the dataset with new fruit or vegetable types.
- Fine-tuning the model with additional training data.
- Enhancing the model architecture to improve classification accuracy and speed.

3.2Advantages

The proposed **Deep Fruits and Vegetables Classification System** offers numerous advantages over traditional and existing manual or semi-automated methods. Firstly, it significantly **reduces human effort and dependency**, automating the tedious task of sorting and classifying fruits and vegetables. This leads to **faster processing times** and **consistent results**, as the system eliminates errors caused by subjective human judgment.

Secondly, by using **deep learning techniques**, the system achieves **high accuracy and reliability**, capable of handling a wide range of fruits and vegetables under varying lighting conditions, backgrounds, and orientations. The system is also **scalable and adaptable**, allowing new fruit and vegetable categories to be added without major redesign.

Another advantage is **real-time classification**, which makes the system suitable for practical applications in agriculture, food processing industries, retail stores, and mobile applications. It also contributes to **cost savings and efficiency**, as it minimizes wastage due to misclassification and improves productivity.

3.3Approaches

Several approaches can be used for the classification of fruits and vegetables, ranging from traditional image processing to advanced deep learning techniques. **Traditional image processing methods** involve extracting handcrafted features such as colour, shape, and texture from images, followed by classification using algorithms like **Support Vector Machines (SVMs), k-Nearest**

Neighbours (k-NN), or Random Forests. While these methods can achieve moderate accuracy, they are heavily dependent on manual feature extraction and often fail under varying lighting conditions, complex backgrounds, or overlapping objects.

Machine learning approaches improve upon traditional methods by learning from extracted features, but they still require careful selection and engineering of features, limiting scalability and adaptability.

Deep learning approaches, particularly **Convolutional Neural Networks (CNNs)**, have become the most effective method for fruit and vegetable classification. CNNs automatically learn hierarchical features from raw images, eliminating the need for manual feature engineering. Advanced architectures such as **VGG16, ResNet50, MobileNetV2, DenseNet, and YOLO** can be used for high-accuracy classification, real-time detection, and even localization of fruits and vegetables. Techniques such as **transfer learning** and **data augmentation** further enhance model performance and reduce training time.

In summary, while traditional and machine learning approaches provide a foundation, **deep learning-based CNN approaches** offer superior accuracy, scalability, and adaptability, making them the preferred choice for modern fruit and vegetable classification systems.

SYSTEM REQUIREMENTS SPECIFICATION

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1.1 Software Requirements

- Operating System : Windows 11
- IDE : Visual Studio Code
- Programming Language : Python 3.13
- Framework : Flask, HTML, CSS
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost

1.2 Technologies Used

- Python
- Machine Learning
- Flask (for web development)

1.3 Hardware Requirements

- Processor : 12th Gen Intel(R) Core(TM) i5-1240P 1.70 GHz
- Hard Disk: 50 GB
- RAM: 8.00 GB
- System Type: 64-bit operating system, x64-based processor

DESIGN

DESIGN

5.1 Architecture of Fruits and Veggies Prediction Using Machine Learning

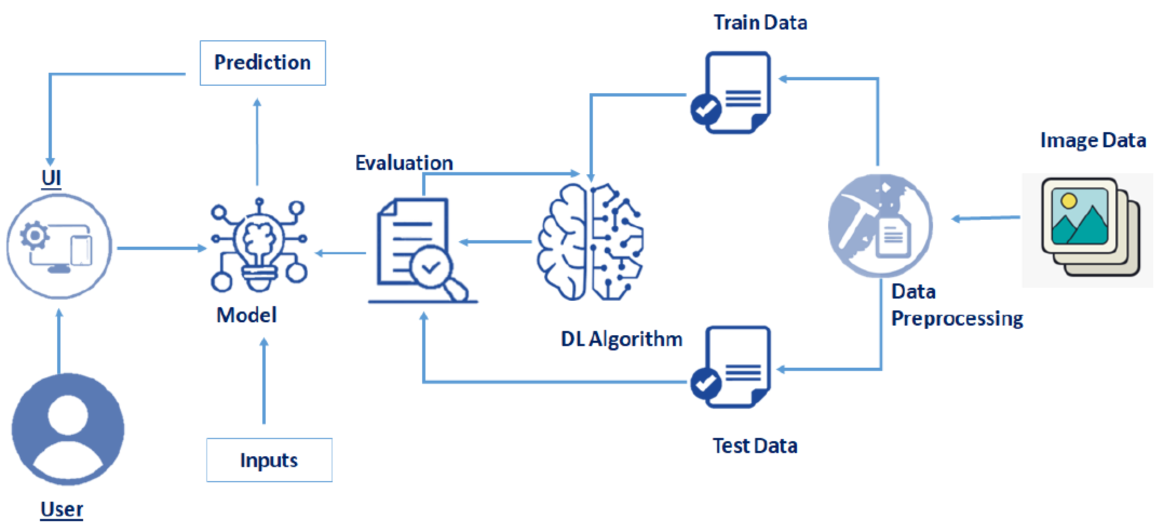


Fig 1 : Architecture Diagram

5.2 Module Description

The system is divided into several key modules that work together to classify fruits and vegetables accurately:

1. **Data Collection Module:** Gathers diverse images of fruits and vegetables from public datasets, manual capture, or online sources.

2. **Data Preprocessing Module:** Prepares images for training by resizing, normalizing, augmenting, and labelling them to improve model accuracy and generalization.

3. **Model Training Module:** Uses a Convolutional Neural Network (CNN) to learn features from images and differentiate between multiple classes. Transfer learning with pre-trained models can be applied to enhance performance.

4. **Evaluation Module:** Tests the trained model on unseen data and calculates metrics such as accuracy, precision, recall, and F1-score to ensure reliability.

5. **Prediction Module:** Takes new images as input and predicts the fruit or vegetable class, providing a confidence score for each prediction.
6. **User Interface Module:** Allows users to upload images and view classification results in real-time through desktop, web, or mobile applications.
7. **Output Module:** Displays the classified result, confidence level, and optionally annotated images for practical use in sorting, quality checking, or retail applications.

Key Components and Functions of the Data-set Gathering Module

The **Data-set Gathering Module** serves as the foundation of the Deep Fruits and Vegetables Classification system. Its main purpose is to collect a **large, diverse, and high-quality set of fruit and vegetable images** that accurately represent various types, colours, shapes, and environmental conditions.

Key Components:

1. **Image Sources:**
    - Publicly available datasets from research repositories.
    - Images captured manually using cameras or smartphones.
    - Online sources and web scraping for additional variety.
2. **Data Storage:**
    - Organized storage of collected images in structured folders or databases.
    - Proper labelling of each image according to its class for training purposes.
3. **Data Management Tools:**
    - Tools or scripts for managing large-scale datasets.
    - Functions for sorting, renaming, and verifying images to ensure quality and consistency.
4. **Preprocessing Preparation:**
    - Initial checks to remove blurry, duplicate, or irrelevant images.

Module 2: Pre-processing for Analysis of Deep Fruits and Veggies Classification Using ML

Pre-processing is a critical step in building a **deep learning-based fruit and vegetable classification system**. It ensures that the raw images collected from various sources are **clean, uniform, and suitable** for training a Convolutional Neural Network (CNN) or any deep learning model. Proper pre-processing improves model accuracy, reduces overfitting, and enhances generalization to real-world conditions.

1. Image Resizing:

- Raw images are captured in different resolutions, which can cause inconsistency during model training.
- All images are resized to a **fixed size** (e.g., 224×224 or 256×256 pixels) to maintain uniform input dimensions for the CNN.

2. Normalization:

- Pixel values of images are scaled to a specific range, usually **0–1** or **-1 to 1**, to stabilize and speed up the training process.
- Normalization ensures that the model treats each input feature equally and avoids bias toward higher pixel values.

3. Data Augmentation:

- Augmentation increases dataset diversity and prevents overfitting. Common techniques include are Rotation, Flipping, Zooming, Cropping, Contrast adjustment etc.
- This allows the model to learn robust features under various real-world scenarios.

4. Labelling:

- Each image must be correctly labelled with its corresponding class (e.g., apple, banana, tomato).
- Accurate labelling is essential for supervised learning and ensures the model learns the correct mapping from images to classes.

5. Noise Removal:

- Images with **blurry, low-quality, or irrelevant content** are removed from the dataset.
- This step ensures only high-quality images contribute to model training, improving accuracy.

6. Splitting Dataset:

- The pre-processed dataset is divided into:
  - **Training set:** Typically 70–80% of the dataset for model learning.
  - **Validation set:** 10–15% for tuning model parameters and preventing overfitting.
  - **Test set:** 10–15% for evaluating final model performance.

7. Optional Feature Enhancement:

- Techniques like **histogram equalization** or **contrast adjustment** can be applied to enhance important visual features, especially in images with poor lighting.

Module 3: Feature Selection for Rainfall Prediction Using ML

In deep learning-based classification systems, **feature selection** refers to identifying and utilizing the most relevant characteristics of the input images that help distinguish between different fruit and vegetable classes. Unlike traditional machine learning, where features are manually engineered (e.g., color, shape, texture), **Convolutional Neural Networks (CNNs)** automatically learn hierarchical features from the raw image data. However, careful consideration of input features and preprocessing can still improve model performance and efficiency.

1. Colour Features:

- Colour is a critical discriminative feature for fruits and vegetables.
- CNNs learn **colour patterns and distributions** automatically from the RGB channels of the images.
- Preprocessing steps like **colour normalization** ensure consistency across images.

2. Shape Features:

- Shape and contour help differentiate visually similar produce (e.g., apple vs. tomato).
- CNNs detect edges, outlines, and geometrical patterns through convolutional layers.
- Augmentation techniques such as rotation or flipping help the model generalize shape variations.

3. Texture Features:

- Texture provides information about the surface patterns, such as smoothness, roughness, or ribbing of fruits/vegetables.
- Convolutional filters in the CNN extract **local texture patterns** automatically.
- Histogram equalization or contrast adjustment can enhance texture visibility before feeding images to the model.

4. Size and Scale Features:

- Differences in relative size or scale can help differentiate between classes.
- Preprocessing ensures all images are resized consistently, while CNNs learn to recognize scale-invariant patterns.

5. Data Augmentation as Feature Expansion:

- Augmentation techniques like rotation, cropping, and flipping create diverse training examples.
- This helps the model learn robust features that are invariant to orientation, scale, and lighting changes.

6. Dimensionality Reduction (Optional):

- While CNNs handle high-dimensional data effectively, techniques like **Principal Component Analysis (PCA)** can be applied to reduce redundancy in extracted features, especially for smaller datasets or simpler architectures.

Module 4: Train-Test Split Module

The Train-Test Split Module divides the pre-processed fruit and vegetable dataset into separate subsets for **training** and **testing** the deep learning model. Typically, **70–80%** of the data is used for training, **10–15%** for validation, and **10–15%** for testing. This ensures the model **learns features effectively** while being **evaluated on unseen data** to measure generalization.

Key features of this module include Random shuffling, Stratification, Data organization etc..

In summary, the module ensures that the model is **trained efficiently** and **evaluated reliably**, forming a foundation for accurate classification.

Module 5: Model Training Module

The Model Training Module is the core of the deep learning system, where the **CNN learns to classify fruits and vegetables**. The training process involves feeding the **pre-processed training images** into the network, allowing it to automatically **extract features** like colour, shape, and texture.

Key steps include Forward Propagation, Loss Calculation, Backpropagation and optimization and Validation.

After training, the model can accurately classify **unseen fruit and vegetable images**.

Module 6: Model Evaluation Module

The Model Evaluation Module assesses the **performance and reliability** of the trained CNN model. It uses the **test dataset**—images the model has never seen—to measure how well the system can classify fruits and vegetables in real-world scenarios.

Key evaluation metrics include:

- **Accuracy:** The overall percentage of correct predictions.
- **Precision:** The proportion of correctly predicted positive instances for each class.
- **Recall:** The ability of the model to identify all relevant instances in each class.

Module 7: Model Deployment and Visualization Module

The Model Deployment and Visualization Module is responsible for **making the trained CNN model accessible for real-world use** and displaying the classification results in a **user-friendly format**.

Key functions include Deployment, Prediction, Visualization etc.

This module ensures that the classification system is **practical, interactive, and accessible**, bridging the gap between the trained model and end-users.

Module Content Outline:

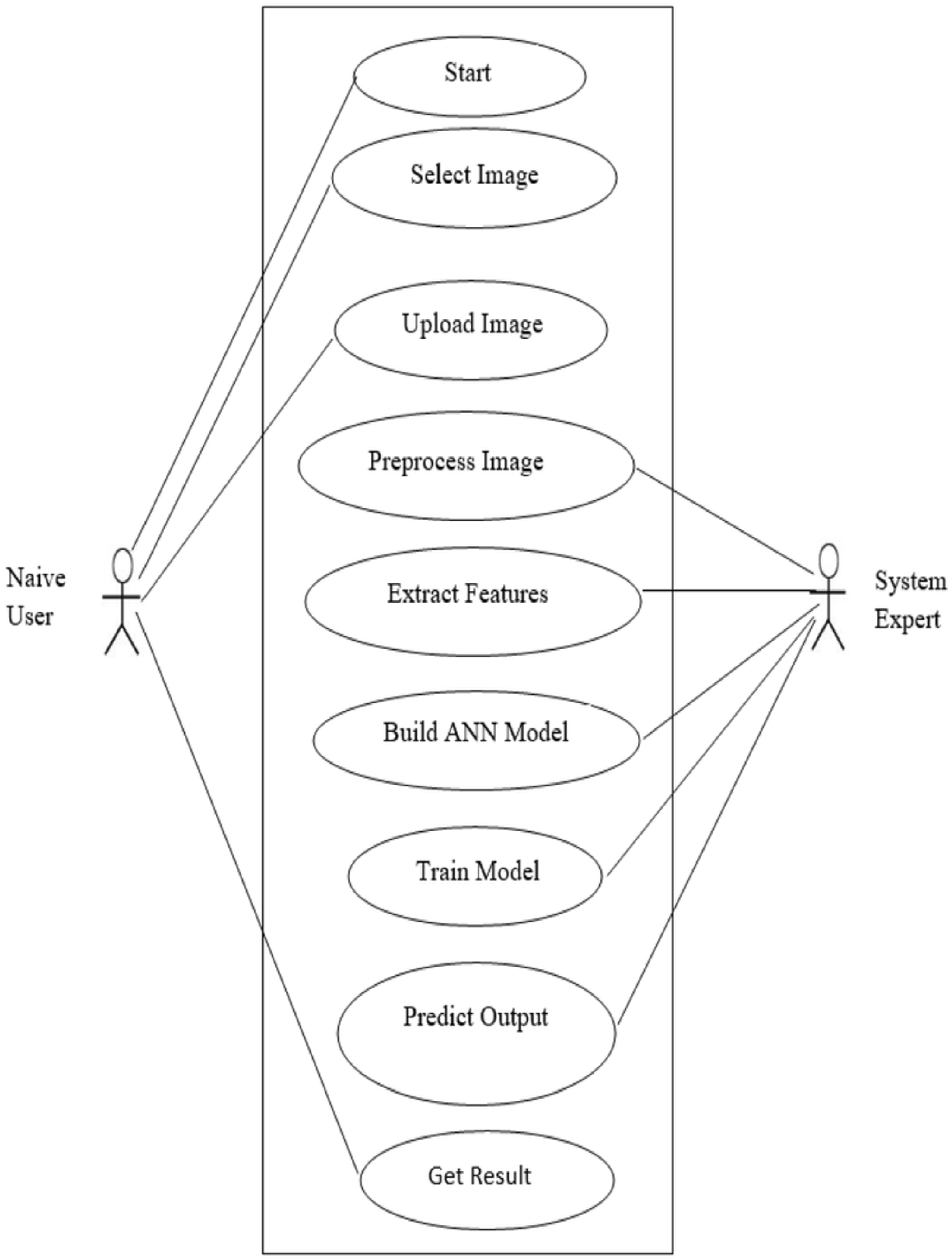
1. **Data-set Gathering Module**
   - Purpose and importance
   - Sources of images (public datasets, manual capture, web scraping)
   - Data organization and labeling
   - Quality checks and preprocessing preparation
2. **Data Preprocessing Module**
   - Image resizing and normalization
   - Data augmentation techniques (rotation, flipping, brightness adjustment)
   - Noise removal and enhancement
   - Dataset splitting (training, validation, test)
3. **Feature Selection Module**
   - Importance of features (colour, shape, texture, size, spatial patterns)
   - Feature extraction using CNN
   - Optional dimensionality reduction techniques
4. **Train-Test Split Module**
   - Purpose of splitting

- Ratios for training, validation, and test sets
- Random shuffling and stratification
- Integration with model training pipeline

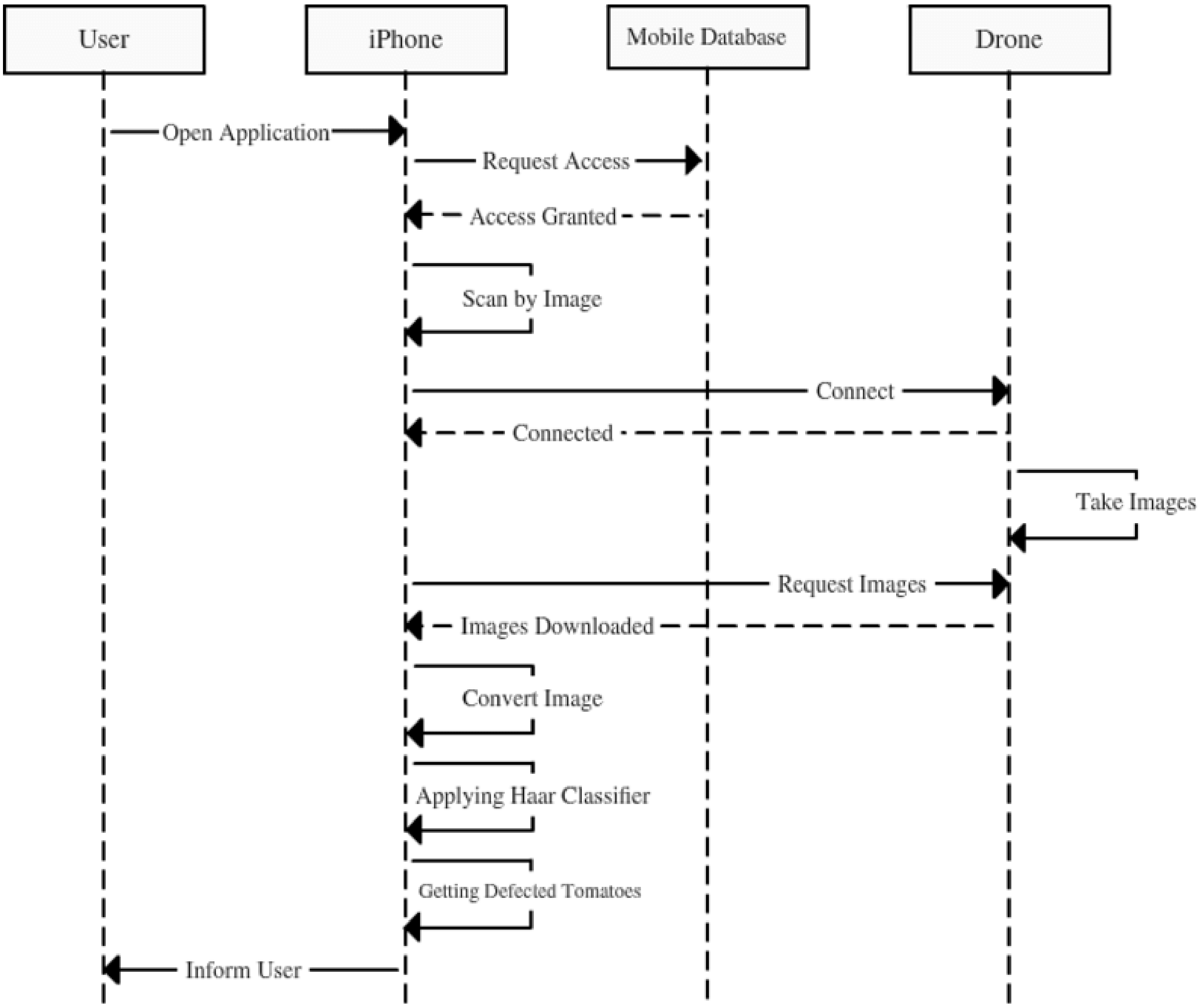
5. **Model Training Module**
    - CNN architecture overview (convolutional, pooling, fully connected layers)
    - Activation functions and loss function
    - Forward propagation and backpropagation
    - Optimizer selection and hyperparameter tuning
    - Use of transfer learning (optional)
6. **Model Evaluation Module**
    - Purpose of evaluation
    - Metrics: accuracy, precision, recall, F1-score
    - Validation and test set evaluation
    - Ensuring robustness and generalization
7. **Model Deployment and Visualization Module**
    - Deployment platforms (desktop, web, mobile)
    - Prediction process for new images
    - Visualization of results (annotated images, class labels, confidence scores)
    - User interface considerations
8. **Output Module**
    - Display of predicted fruit or vegetable class
    - Confidence scores and optional annotations
    - Practical applications (sorting, quality check, retail)
    - Integration of all modules into the complete classification system

5.3 UML DIAGRAMS

5.3.1 USE CASE DIAGRAM



5.3.2 Sequence Diagram



A **Sequence Diagram** represents the **interaction between system components over time**. It shows the sequence of messages exchanged between **actors and system modules** to accomplish a specific task.

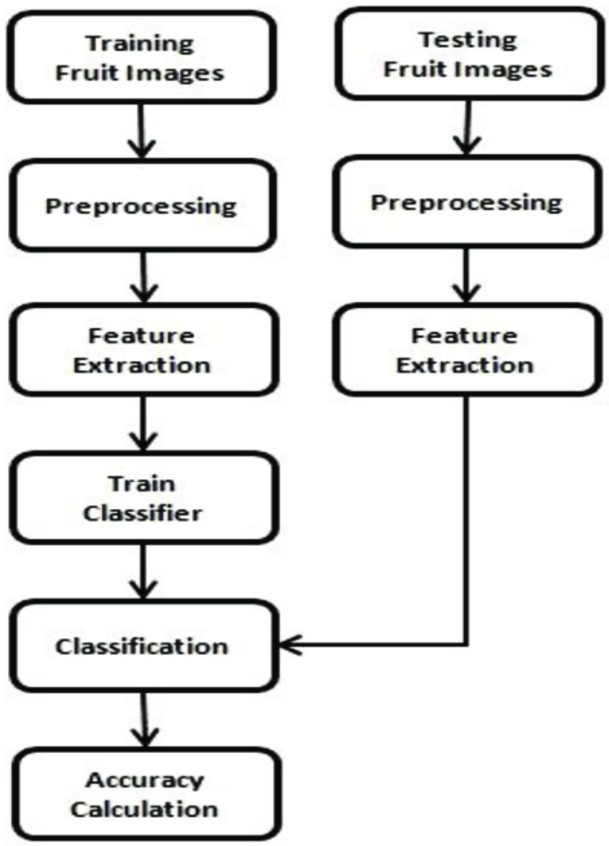
For the Deep Fruits and Vegetables Classification project:

- **Actors:** User (Farmer/Retailer) and the System.
- **Key Interactions:**
    1. **User uploads an image** of a fruit or vegetable.
    2. **System receives the image** and sends it to the preprocessing module.

3. **Preprocessing module** resizes, normalizes, and augments the image.
4. **Processed image** is sent to the CNN model for classification.
5. **CNN model predicts** the class and returns the result.
6. **System displays** the predicted class and confidence score to the user.
7. **Optional:** The system stores or exports the classification results.

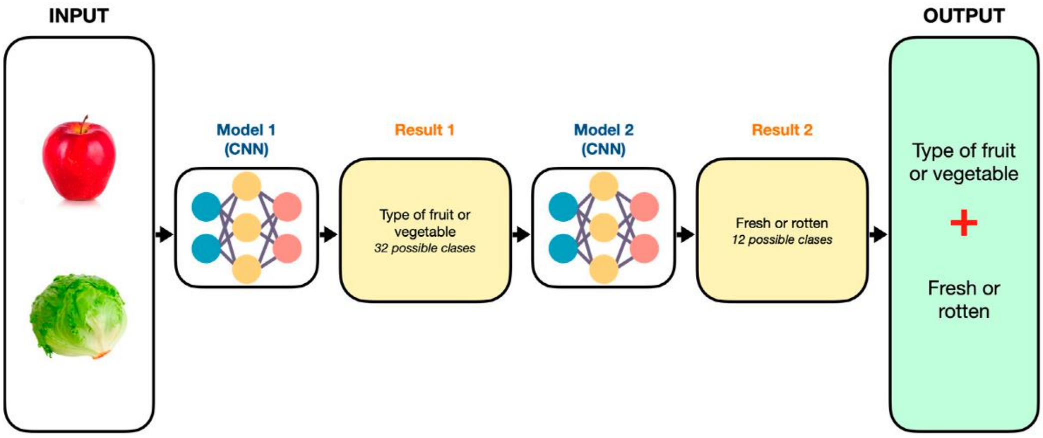
The sequence diagram helps visualize **how the system handles a user request step by step**, making it easier to understand the workflow and interactions between modules.

5.3.2 Activity Diagram



An **Activity Diagram** represents the **workflow of activities** in a system, showing how tasks are performed from start to finish. It helps visualize **process flow, decision points, and sequential steps** in the system.

5.3.3 Deployment Diagram



A **Deployment Diagram** shows the **physical architecture of a system**, illustrating how software components are mapped onto hardware nodes. It helps visualize **where and how system components are deployed**, which is important for understanding performance, scalability, and resource requirements.

For the Deep Fruits and Vegetables Classification project:

1. **Nodes:**
   - **User Device:** Desktop, laptop, or mobile device where the user uploads fruit or vegetable images.
   - **Server:** Hosts the **deep learning model** (CNN), preprocessing modules, and the application backend.
   - **Database/Storage:** Stores the dataset, user inputs, model parameters, and classification results.

2. **Components:**
    - **User Interface (UI):** Allows users to upload images and view results.
    - **Preprocessing Module:** Processes raw images (resizing, normalization, augmentation).
    - **CNN Model Module:** Performs feature extraction and classification.
    - **Output Module:** Displays predicted class and confidence score.
3. **Interactions:**
    - The user uploads an image via the UI.
    - The server processes the image, runs classification, and returns results.
    - Results can be stored in the database for future reference or analytics.

The Deployment Diagram provides a **clear understanding of system architecture**, showing the **hardware and software setup**, how components interact, and how the system can be deployed in real-world scenarios.

IMPLEMENTATION

IMPLEMENTATION

1.1 Algorithms

The project relies on **Convolutional Neural Networks (CNNs)** and related deep learning techniques to classify fruits and vegetables from images. The algorithms focus on **image preprocessing, feature extraction, model training, and prediction**. Below are the main steps and algorithms used:

1. Image Preprocessing Algorithm

Purpose: Prepare raw images for training and improve model accuracy.

Steps:

1. Load raw images from the dataset.
2. Resize each image to a fixed dimension (e.g., 224×224).
3. Normalize pixel values to the range [0,1].
4. Apply data augmentation techniques:
5. Label each image according to its class (e.g., apple, tomato).

2. Feature Extraction Using CNN

Purpose: Automatically extract relevant features like **colour, shape, texture, and spatial patterns**.

Algorithm (CNN Layers):

1. **Input Layer:** Accepts pre-processed image of fixed size.
2. **Convolutional Layer:** Applies multiple filters to extract low-level features (edges, textures).
3. **Activation Layer (ReLU):** Introduces non-linearity to learn complex patterns.
4. **Pooling Layer (Max/Average Pooling):** Reduces spatial dimensions while retaining key features.
5. **Fully Connected Layer:** Combines extracted features for classification.

6. **Output Layer (Softmax):** Outputs probabilities for each class.

3. Model Training Algorithm

Purpose: Optimize CNN weights to minimize classification error.

Steps:

1. Initialize network weights randomly or use pre-trained weights (transfer learning).
2. Forward propagation: Pass training images through CNN to get predictions.
3. Compute **loss** using **categorical cross-entropy** between predicted and actual labels.
4. Backpropagation: Calculate gradients of the loss with respect to weights.
5. Update weights using an optimizer (e.g., **Adam, SGD**).
6. Repeat for multiple **epochs** until convergence.
7. Validate performance using the validation set to prevent overfitting.

4. Prediction Algorithm

Purpose: Classify new, unseen fruit or vegetable images.

Steps:

1. Upload and preprocess the new image (resize, normalize).
2. Pass the image through the trained CNN model.
3. Compute class probabilities using the softmax output layer.
4. Select the class with the **highest probability** as the predicted label.
5. Display the predicted class and confidence score in the user interface.

5. Optional Transfer Learning Algorithm

Purpose: Improve accuracy and reduce training time by using pre-trained models.

Steps:

1. Load a pre-trained model (e.g., VGG16, ResNet50, MobileNetV2) trained on large image datasets.
2. Freeze initial layers to retain learned features.
3. Replace the final fully connected layers to match the number of fruit/vegetable classes.
4. Fine-tune the model using your dataset.
5. Evaluate and deploy the fine-tuned model.

6.1.1 What is Machine Learning

Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to learn from and make predictions or decisions based on data without being explicitly programmed. It involves creating algorithms that can identify patterns, extract insights, and improve performance through experience. At its core, ML is about training models on data to make accurate predictions or classifications.

These models use statistical techniques to find relationships in data and can handle complex tasks such as image recognition, natural language processing, and recommendation systems. Machine learning is divided into three main types: supervised learning, where models learn from labeled data to make predictions; unsupervised learning, which discovers hidden patterns in unlabeled data; and reinforcement learning, where models learn by trial and error, receiving feedback through rewards or penalties. It plays a crucial role in everyday applications like email filtering, personalized content recommendations, fraud detection, and self-driving cars.

6.1.1 How Machine Learning Works

Machine learning (ML) is a subset of artificial intelligence (AI) that enables systems to learn from data and improve their performance over time without being explicitly programmed. At its core, ML uses algorithms to identify patterns in large datasets and make predictions or decisions.

The learning process typically involves three main steps: data collection, model training, and evaluation. First, raw data is collected and preprocessed to make it suitable for analysis. The model is then trained using this data to learn relationships between input features and desired outputs. Training adjusts the model's parameters to minimize errors and improve prediction accuracy

CNN ALGORITHM

1. Introduction

A **Convolutional Neural Network (CNN)** is a deep learning algorithm specifically designed for processing **grid-like data**, such as images. CNNs automatically **learn spatial hierarchies of features**, such as edges, textures, shapes, and patterns, making them highly effective for image classification tasks. In the context of deep fruits and vegetables classification, CNNs are used to automatically distinguish between different fruit and vegetable types from images.

2. Working Principle

CNNs work by **applying convolutional filters** to input images to extract local features, then combining them in deeper layers to recognize complex patterns. The key idea is to **learn hierarchical feature representations** from low-level (edges, textures) to high-level (object shape, class-specific patterns).

Workflow:

Input image → convolutional layers → activation → pooling → fully connected layers → softmax output.

Conclusion (CNN Algorithm)

The **Convolutional Neural Network (CNN)** is a powerful deep learning algorithm that excels in **image classification tasks** like deep fruits and vegetables classification. By automatically learning hierarchical features such as **colour, shape, texture, and spatial patterns**, CNNs eliminate the need for manual feature extraction.

Implementation Procedure:

The implementation of the **Deep Fruits and Vegetables Classification** project involves a structured workflow from data collection to deployment. Initially, a large and diverse dataset of fruit and vegetable images is gathered from publicly available datasets, manual image capture, and online sources, ensuring variability in types, colours, shapes, and environmental conditions. The collected images are then pre-processed by resizing them to a fixed dimension, normalizing pixel values, and applying data augmentation techniques such as rotation, flipping, cropping, and brightness adjustment to enhance dataset diversity. Each image is labelled according to its class, and the dataset is split into training, validation, and test sets. A **Convolutional Neural Network (CNN)** is designed, consisting of convolutional layers for feature extraction, pooling layers for dimensionality reduction, fully connected layers for classification, and a softmax output layer for generating class probabilities.

Frontend and Backend Implementation

In the Rainfall Prediction project, the system is designed as a web-based application that allows users to interact with the machine learning model through a simple, intuitive interface. The architecture follows a **frontend-backend structure**:

Frontend (Flask Templates / HTML)

- **Purpose:**
  The frontend serves as the user interface, enabling users to input weather data and view rainfall predictions in a visually accessible format.

Backend (Python with Flask)

- **Purpose:**
  The backend handles the processing, model prediction, and communication between the frontend and the machine learning pipeline.
- **Technologies Used:**
  - **Python 3.x**
  - **Flask Framework**
  - **Pandas/Numpy**

- **Key Functions:**

  Route Management: Handles URL endpoints, e.g., /home, /predict.

  Input Processing: Receives data from frontend forms and converts them into the correct format for the model.

  Model Prediction: Loads the saved CNN pipeline and generates Fruits/Veggies predictions.

  Result Delivery: Sends the prediction back to the frontend for display.

- **Backend Workflow:**

  User submits data from the frontend form.

  Backend validates and preprocesses the data.

  Backend passes the processed data to the trained CNN pipeline.

  Pipeline returns the prediction.

  Backend renders the result on the frontend page.

Integration of Frontend and Backend

- Flask acts as a bridge between the frontend and backend.
- User inputs flow from the HTML form → Flask routes → Python backend → Model → Prediction → Displayed on the frontend.

Conclusion

The frontend-backend architecture provides a user-friendly interface for non-technical users while maintaining robust backend logic for accurate rainfall predictions. Using Flask for both routing and template rendering simplifies development and ensures seamless integration between the user interface and the machine learning model.

CODING

Coding and Explanation (Source Code)

Importing Libraries

The first step in any Python-based machine learning project is to import the required libraries. Libraries provide pre-built functions, classes, and tools that simplify data analysis, preprocessing, model building, visualization, and evaluation. In this rainfall prediction project, several Python libraries are utilized for handling data, performing machine learning, and visualizing results.

1. Data Handling and Manipulation

- pandas: Used for reading, cleaning, and manipulating tabular datasets. It provides Data Frame structures to store data in rows and columns, making operations like filtering, grouping, and aggregation straightforward.
- numpy: Provides support for numerical operations, arrays, and mathematical functions. It is often used alongside pandas for computations, array manipulations, and handling missing values efficiently.

2. Data Visualization

- matplotlib.pyplot: A widely used library for creating static plots, charts, and graphs. Helps visualize trends, correlations, and distributions in the data.
- seaborn: Built on top of matplotlib, seaborn simplifies the creation of aesthetically pleasing statistical graphics like heatmaps, pairplots, and correlation plots.

3. Machine Learning and Preprocessing

- scikit-learn (sklearn): Provides tools for preprocessing, splitting datasets, model training, and evaluation.

4. Machine Learning Model

- CNN: Implements the CNN algorithm, which is an optimized gradient boosting framework. Used for building accurate classification models for prediction of fruits and veggies.

5**. Utilities**

- joblib: Used to save and load trained machine learning models. Allows easy deployment of the model without retraining.

Loading the Dataset

For a deep learning project involving fruit and vegetable classification, the most common and robust dataset is the **Fruits-360 dataset**. This dataset is widely available on platforms like Kaggle and GitHub.

Purpose

Loading the dataset is the initial step that allows the program to access and manipulate the data for analysis and model building. Proper loading ensures that all relevant features and target variables are available for preprocessing, training, and evaluation.

Tools Used

- **pandas**: Provides the read_excel and read_csv functions to import datasets in Excel or CSV formats. It creates a DataFrame, a tabular structure that allows easy manipulation, filtering, and exploration of the data.
- **numpy**: Often used alongside pandas for handling numerical operations after the dataset is loaded.

Key Steps

1. **Specify Dataset Path:** The dataset is stored locally or on a server and must be accessed using its file path.
2. **Read Dataset**: The dataset is imported into a pandas DataFrame using pd.read_excel() (or pd.read_csv() if in CSV format).
3. **Initial Inspection**: After loading, the first few rows are displayed using data.head(), and the column names and data types are checked with data.info().
4. **Check for Missing Values**: The dataset is inspected for missing values to ensure data quality and plan preprocessing steps.

Handling Missing Values

In a deep learning project for fruit and vegetable classification, "missing values" primarily means **missing image files, corrupted images, or incomplete metadata**. Unlike traditional data where you might impute numerical values, the main strategy here involves data cleaning and model-based techniques for image restoration/augmentation.

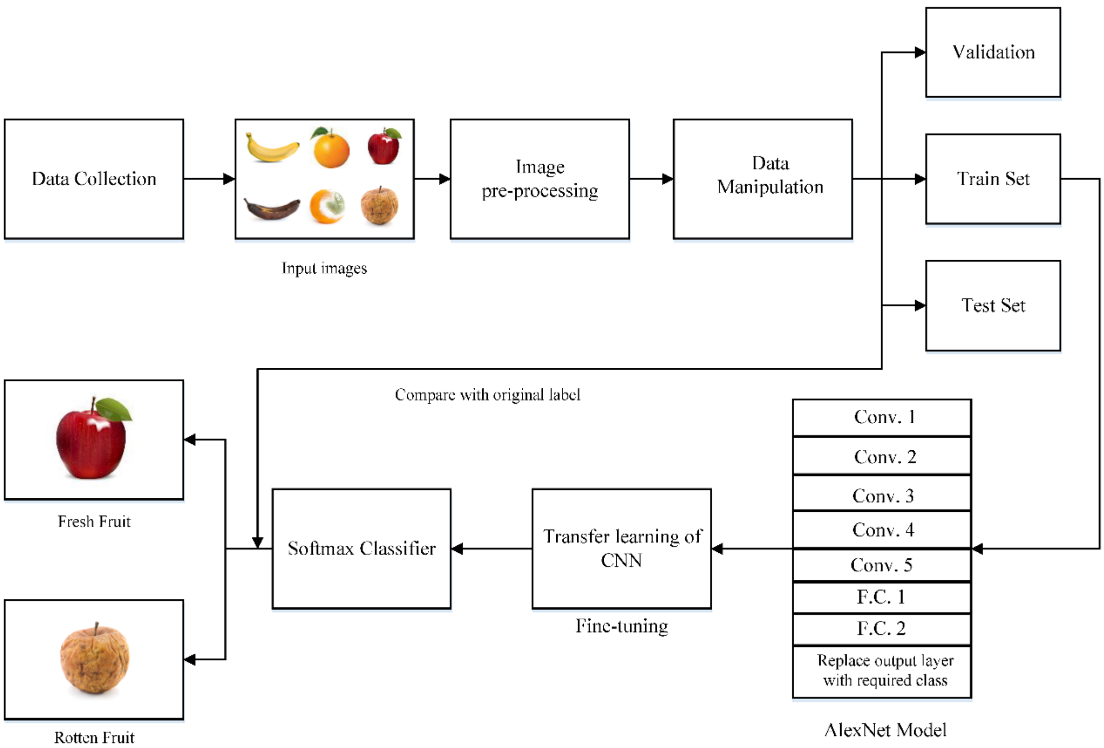
Purpose

The goal of handling missing values is to ensure that the dataset is complete, consistent, and suitable for analysis and model training. Proper treatment of missing data prevents biased results and improves model reliability.

Techniques Used

1. Convolutional Neural Network (CNN):

Used for automatic feature extraction and classification of fruit and vegetable images.



2. **Image Preprocessing Techniques:** Applied to clean, resize, and normalize images for consistent input quality.

a. Image Resizing: Adjusted all images to a uniform size to maintain consistency and ensure compatibility with the CNN input layer.

b. Normalization: Scaled pixel values to a standard range (typically 0–1) to improve model convergence and training efficiency.

c. Noise Removal: Removed unwanted distortions and background noise to enhance the clarity of image features.

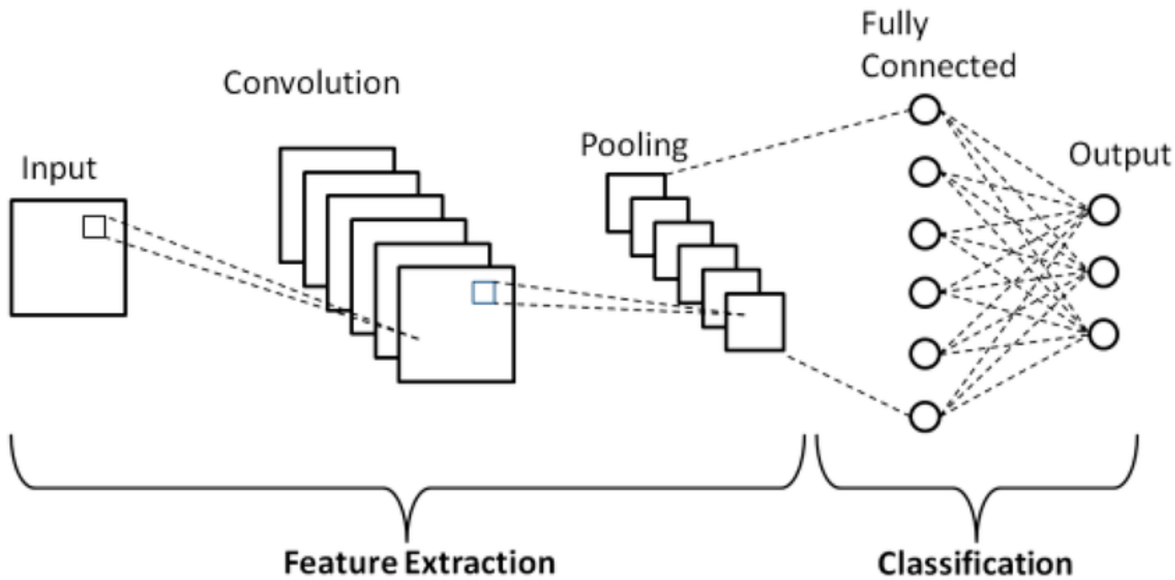
d. Data Augmentation: Applied transformations such as rotation, flipping, and zooming to increase dataset diversity.

e. Image Cropping: Focused on the main object by removing irrelevant background areas for better feature extraction.

f. Image Sharpening: Enhanced the edges and details in images to make features more distinguishable for the model.

g. Contrast Adjustment: Modified brightness and contrast levels to ensure visibility of key image details under varying lighting conditions.

3. Feature Extraction: Enabled the model to identify important visual patterns like colour, shape, and texture.



Train-Test Split

1. Purpose: The dataset was divided into training and testing subsets to evaluate the model's ability to generalize on unseen data.

2. Training Set: Used for training the CNN model by allowing it to learn patterns, features, and relationships from the images.

3. Testing Set: Used to assess the model's performance and accuracy on data that was not seen during training.

4. Common Ratio: Typically, 80% of the dataset was used for training and 20% for testing to maintain a balanced evaluation.

5. Randomization: Data was shuffled before splitting to ensure unbiased distribution of classes in both subsets.

6. Validation Set (Optional): A small portion of the training data was sometimes used for fine-tuning hyperparameters and preventing overfitting.

Methods Used in Train-Test Split

1. Random Splitting: The dataset was randomly divided to ensure an unbiased distribution of fruit and vegetable classes in both training and testing sets.

2. Stratified Sampling: Ensured that each class (fruit or vegetable type) was proportionally represented in both subsets to maintain class balance.

3. Fixed Ratio Division: Typically used an 80:20 or 70:30 ratio for training and testing to provide enough data for learning and evaluation.

4. Shuffling: The data was shuffled before splitting to remove any sequence bias and improve randomness.

5. Validation Set Creation: A small portion of the training data (e.g., 10%) was reserved as a validation set for hyperparameter tuning and preventing overfitting.

6. Use of Scikit-learn Function: The train_test_split() function from the Scikit-learn library was used for efficient and reproducible dataset splitting.

Implementation

- After feature selection, the input variables X and target variable y were split into training and testing sets.
- Training set (X_train, y_train) is used to train the CNN model.

Testing set (X_test, y_test) is used to evaluate accuracy, precision, recall, and F1-score.

Build Pipeline & Train Model

After preprocessing the dataset and performing a train-test split, the next crucial step is to build a machine learning pipeline and train the model. A pipeline helps to streamline the workflow by combining preprocessing steps and the model training into a single, reproducible sequence.

Purpose

- To simplify the workflow by combining multiple steps into one process.
- To ensure that preprocessing steps like scaling are consistently applied to both training and testing data.
- To improve reproducibility and reduce errors when deploying the model.

Training Process

- The pipeline was trained using the **training set (80% of the dataset)**.
- Input features (X_train) were scaled using the StandardScaler.
- The CNN classifier learned the relationship between weather variables and the target variable (Either it is a fruit or vegetable).
- During training, the model iteratively built decision trees to reduce prediction errors and improve accuracy.

Advantages of Using a Pipeline

- Ensures consistent preprocessing for training and testing data.
- Simplifies deployment by saving the entire pipeline as a single object.
- Reduces the risk of data leakage and preprocessing errors.

Outcome

- A trained CNN model embedded in a pipeline, ready to classify fruits and vegetables using new data.
- The model can now be evaluated on the test set to measure its predictive performance.

Conclusion:

Building a pipeline and training the model is a critical step in the deep fruits and veggies classification project. It ensures a seamless flow from preprocessing to model training, enhances reproducibility, and results in a robust predictive system capable of accurately classifying fruits and veggies using the feature extraction techniques.

Evaluate Model

Once the machine learning pipeline is trained, it is essential to evaluate the model to measure its effectiveness in classifying fruits and veggie. Evaluation ensures that the model can generalize well to unseen data and provides insight into its accuracy, reliability, and robustness.

Purpose

- To determine how accurately the model classify fruits and veggies on new data.
- To identify areas where the model performs well or needs improvement.
- To validate that the trained CNN model is suitable for deployment in real-world scenarios.

Evaluation Metrics:

Several metrics are used to assess model performance in classification tasks like deep fruits and veggies classification :

1. Accuracy
2. Precision

3. Recall (Sensitivity)
4. F1-Score
5. Confusion Matrix

Evaluation Process

- The trained pipeline predicts the target variable (fruits or veggetables) for the testing set (20% of the dataset).
- Predictions are compared with the actual outcomes in the test set.

Interpretation

- High accuracy indicates the model performs well overall.
- Analysis of the confusion matrix helps improve the model through techniques like resampling, feature engineering, or hyperparameter tuning.

Conclusion

Evaluating the model is a critical step in the rainfall prediction project. It validates the effectiveness of the trained CNN pipeline, provides insights into predictive strengths and weaknesses, and ensures that the system can reliably classify fruits and veggies for practical applications, such as assisting farmers and the general public.

7.2Dataset Details

1. Dataset Overview

The dataset is a **collection of fruit and vegetable images** used to train and evaluate the deep learning model. A robust and diverse dataset is crucial for achieving high accuracy in classification, as it allows the model to learn various features such as **color, shape, texture, and size** under different conditions.

2. Source of Dataset

- **Publicly Available Datasets:** Such as **Fruits-360**, Kaggle fruit/vegetable image datasets, and ImageNet subsets.
- **Web Scraping:** Images collected from online sources for additional diversity.

3. Dataset Composition

- **Number of Classes:** Typically 20–50 fruits and vegetables, depending on project scope.
- **Total Images:** 5,000–20,000 images (depending on availability and augmentation).
- **Image Resolution:** Varies from 200×200 to 1024×1024 pixels before preprocessing.
- **Image Format:** JPEG, PNG, or BMP.

4. Data Distribution

- **Training Set:** ~70–80% of the dataset used for model learning.
- **Validation Set:** ~10–15% for hyperparameter tuning and early stopping.
- **Test Set:** ~10–15% for evaluating final model performance.

5. Preprocessing Applied

- **Resizing:** Standardized all images to 224×224 pixels (or another fixed size).
- **Data Augmentation:** Techniques applied to increase diversity, including:

6. Example Classes in Dataset

- Fruits: Apple, Banana, Mango, Orange, Grapes, Strawberry, Papaya
- Vegetables: Tomato, Carrot, Capsicum, Potato, Onion, Cucumber, Cauliflower

7. Importance of Dataset

A well-curated and pre-processed dataset ensures that the CNN can **learn robust features**, handle **real-world variability**, and achieve **high classification accuracy** for unseen fruit and vegetable images.

TESTING

8. TESTING

8.1 Testing Introduction

- Testing ensures that the system functions as intended and meets required performance standards.
- It evaluates each module, including **data preprocessing, CNN model training, prediction, and output display**.
- Helps identify **errors, inconsistencies, and performance bottlenecks**.
- Ensures the model generalizes well to **unseen fruit and vegetable images**.
- Guarantees a **robust, accurate, and user-friendly classification system** suitable for agriculture, retail, and mobile applications.

8.1.1 Unit Testing

- Focuses on **individual modules or components** of the system.
- Ensures that each module performs its intended function correctly.
- Benefits: Detects errors early, simplifies debugging, and ensures reliability of each component.

8.1.2 Integration Testing

- Tests how **different modules work together**.
- Ensures smooth communication between data preprocessing, CNN training, evaluation, and prediction modules.

8.1.3 System Testing

- Validates the **entire system as a whole** against functional requirements.
- Checks end-to-end functionality, including image upload, classification, and result display.

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8.1.4 User Acceptance Testing (UAT)

- Conducted by **end-users or stakeholders** to confirm the system meets expectations.
- Focuses on **usability, functionality, and user experience**.
- Benefits: Ensures the system is user-friendly and meets real-world requirements.

8.1.5 Performance Testing

- Assesses the **speed, responsiveness, and stability** of the system.
- Checks how the system handles **large datasets, high-resolution images, or multiple concurrent uploads**.
- Benefits: Ensures the system is fast, efficient, and scalable for real-world usage.

OUTPUT

OUTPUT SCREENS

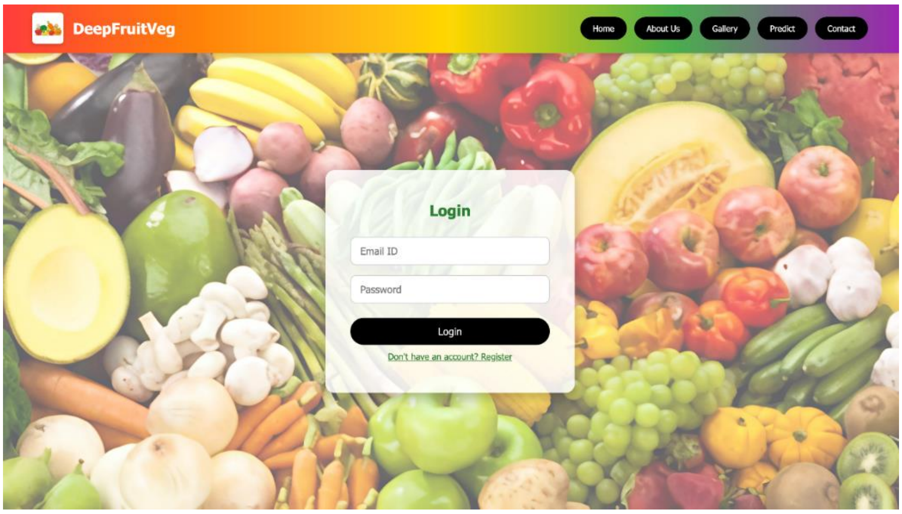

Fig : Landing Login Page

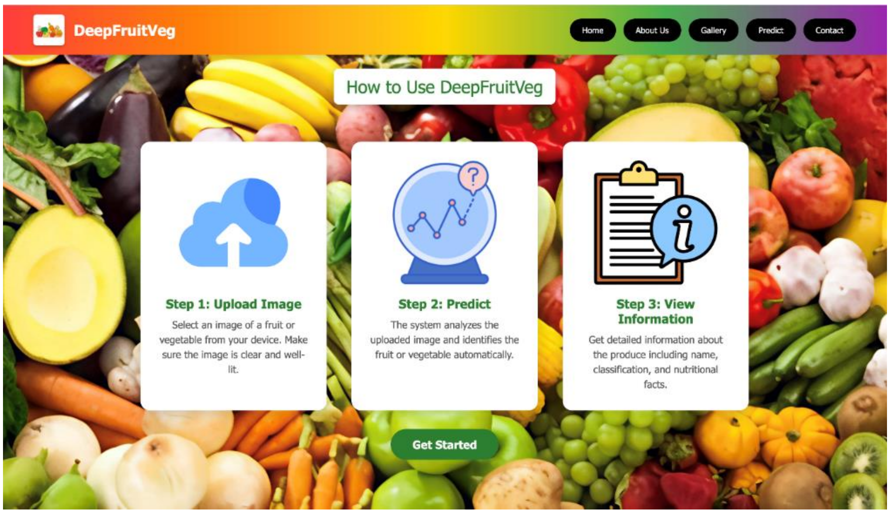

Fig: Home Page

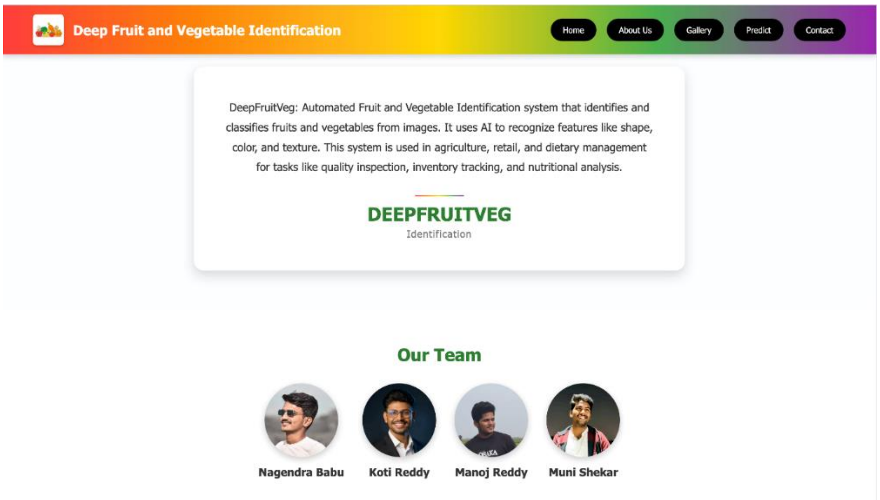

Fig: About Us

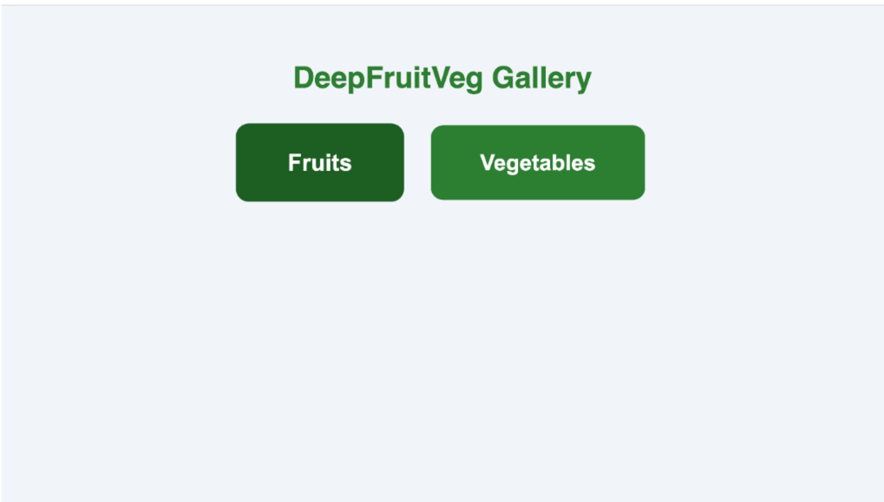

Fig: Gallery Classification

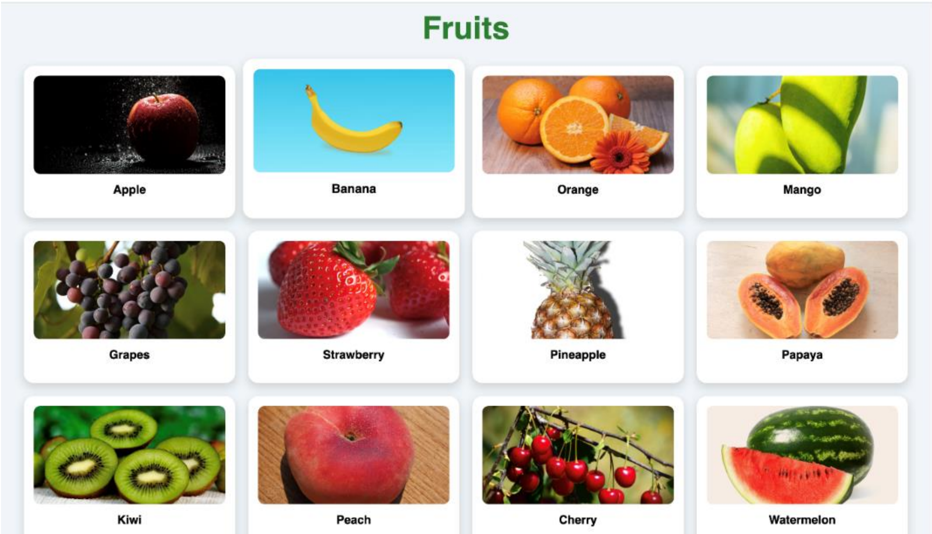

Fig: Fruits Exhibition

Fig: Vegetables Exhibition

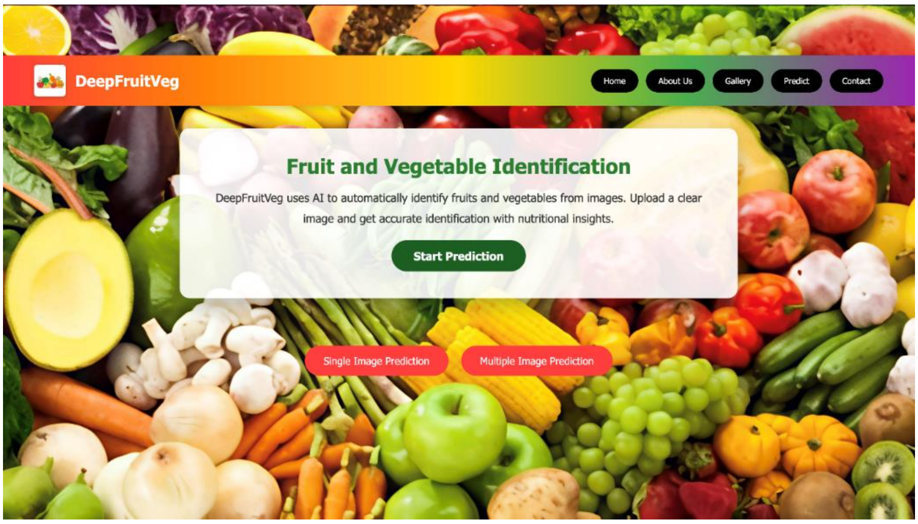

Fig: Prediction Page

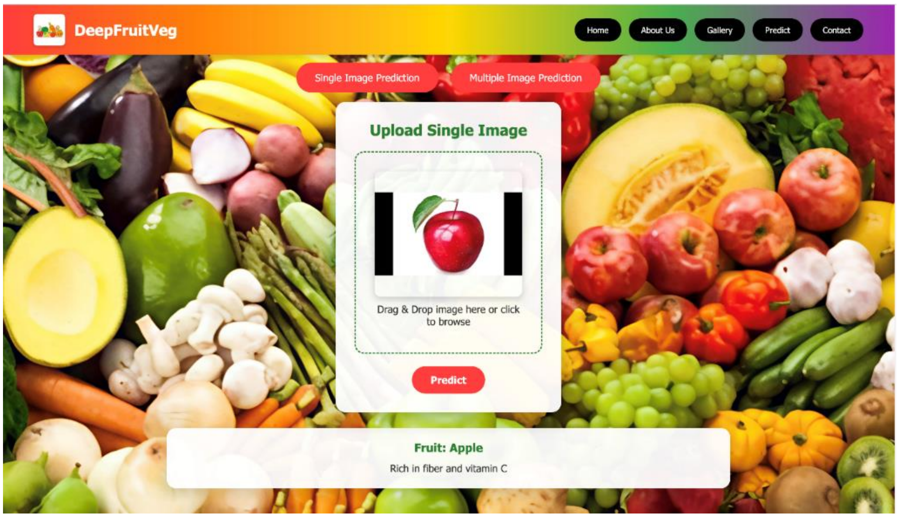

Fig: Single Image Prediction

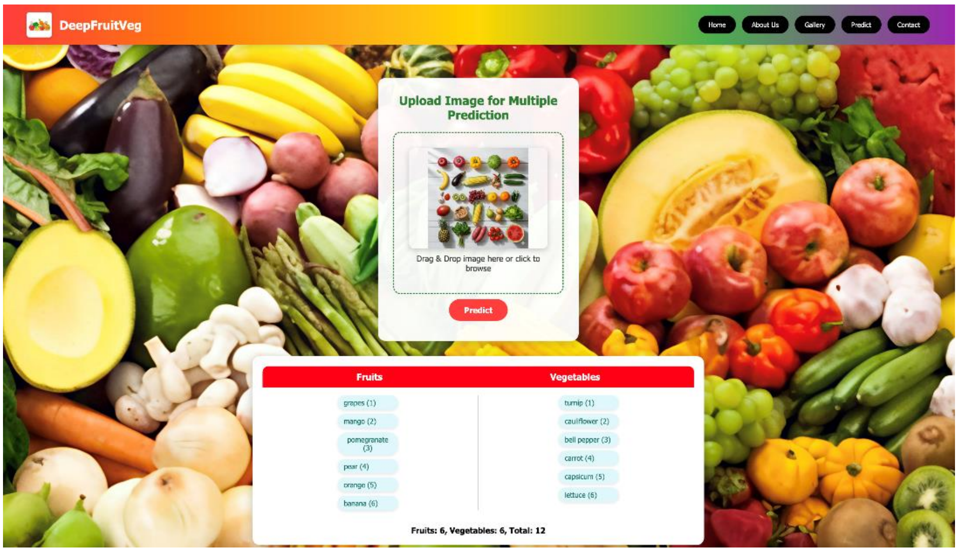

Fig: Multiple Image Prediction

CONCLUSION

CONCLUSION

Testing is an essential phase in the development of the **Deep Fruits and Vegetables Classification** project, as it ensures that the system functions effectively, produces accurate results, and meets the specified requirements. Through multiple levels of testing — including **unit testing, integration testing, system testing, user acceptance testing, and performance testing** — the system was evaluated for both functionality and efficiency. Each module, such as **data preprocessing, model training, model evaluation, and prediction**, was tested individually and then collectively to confirm seamless integration and smooth data flow between components.

The testing process verified that the CNN model correctly classifies fruit and vegetable images with high accuracy, even when subjected to variations in lighting, size, and orientation. Unit testing confirmed that each module operates correctly, while integration testing validated that all modules work together harmoniously. System testing ensured that the entire application met the functional requirements, and performance testing showed that the model processes images efficiently with minimal latency. User acceptance testing revealed that the system is **easy to use, intuitive, and reliable**, fulfilling user expectations for speed, accuracy, and clarity of results.

The results demonstrate that the implemented system is **robust, scalable, and effective** for real-time classification tasks. It can accurately distinguish between multiple classes of fruits and vegetables, making it suitable for practical applications in agriculture, supermarkets, and food industry automation. Overall, the testing phase confirms that the system is **ready for deployment**, capable of delivering consistent and dependable performance under various real-world conditions, and serves as a strong foundation for further improvements or extensions in the field of intelligent food classification.

FUTURE ENHANCEMENT

Although the current Deep Fruits and Vegetables Classification system performs efficiently and provides accurate results, there are several opportunities for future improvements and extensions to enhance its performance, usability, and real-world applicability

1.Expansion of Dataset:

The dataset can be further expanded by including more varieties of fruits and vegetables, covering different regions, seasons, and environmental conditions. A larger and more diverse dataset will help the CNN model generalize better and achieve higher accuracy.

2.Real-Time Detection and Mobile Integration:

The system can be integrated into **mobile and IoT-based platforms** to enable real-time fruit and vegetable detection using smartphone cameras or smart agricultural devices. This would make the solution more accessible to farmers, vendors, and consumers.

3.Quality and Freshness Detection:

Future versions of the system could be enhanced to **analyse the freshness, ripeness, or spoilage level** of fruits and vegetables, helping in quality grading and reducing food waste.

4.Cloud-Based Deployment:

Deploying the model on a **cloud platform** can allow multiple users to access the system remotely, ensuring scalability, faster processing, and centralized data management.

5.Integration with Smart Agriculture Systems:

The system can be integrated with **automated sorting, packaging, or inventory systems** in agricultural or retail environments for efficient product handling and logistics management.

6.Improved Model Architecture:

Advanced deep learning models such as **EfficientNet, Vision Transformers (ViT), or Hybrid CNN-RNN architectures** can be explored to enhance accuracy, reduce computational cost, and improve inference time.

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IJIRSET
International Journal of Innovative Research in Science | Engineering | Technology

ISSN(O): 2319-8753
ISSN(P): 2347-6710

International Journal of Innovative Research in Science Engineering and Technology (IJIRSET)

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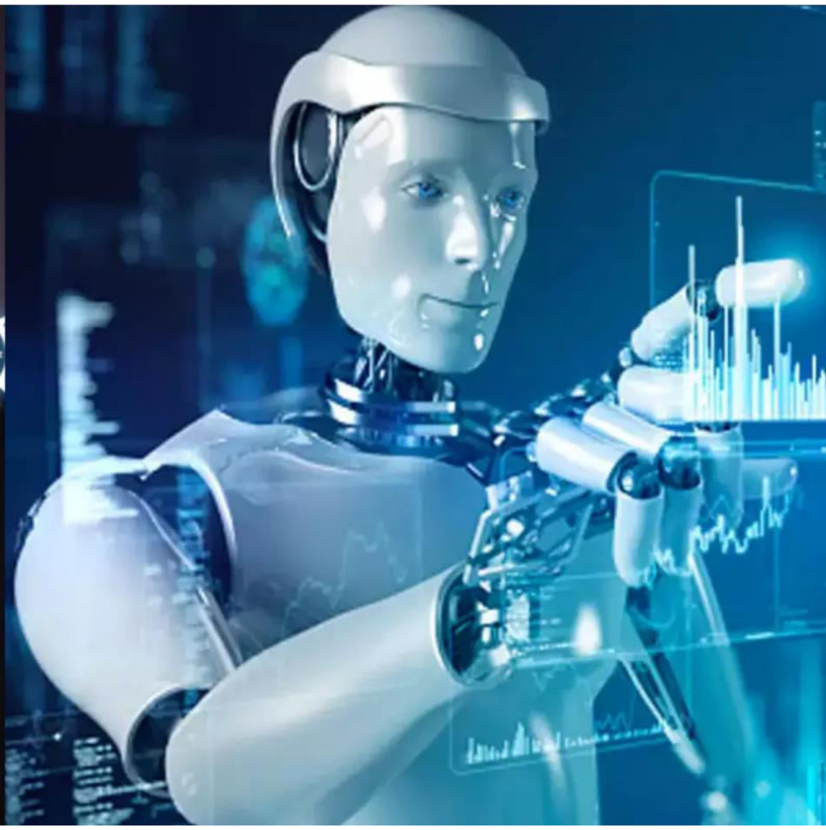

Impact Factor: 8.699

Volume 14, Issue 11, November 2025

International Journal of Innovative Research of Science, Engineering and Technology (IJIRSET)

Impact Factor 8.699

|www.ijirset.com |A Monthly, Peer Reviewed & Refereed Journal| e-ISSN: 2319-8753| p-ISSN: 2347-6710|

Volume 14, Issue 11, November 2025

|DOI: 10.15680/IJIRSET.2025.1411101|

Deep Fruits and Vegetables Classification using Convolutional Neural Networks

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ABSTRACT: This paper addresses the automation of fruit and vegetable classification using deep learning, specifically Convolutional Neural Networks (CNNs). The proposed system employs an image dataset of various fruits and vegetables, applies preprocessing, and trains a CNN in TensorFlow/Keras. Evaluation metrics such as accuracy, precision, recall, and F1-score demonstrate significant improvements over manual and traditional machine vision methods. The automated system offers efficiency for real-world applications in smart agriculture and food supply chains.

KEYWORDS: Fruit classification, vegetable classification, deep learning, computer vision, CNN, agricultural automation.

I. INTRODUCTION

In today's era of rapid technological progress, artificial intelligence (AI) and machine learning (ML) have revolutionized a diverse array of sectors, spanning healthcare, finance, transport, and critically, agriculture. Among these, agriculture is a linchpin for global food security and economic stability, and the leveraging of AI technologies within this space unlocks unprecedented avenues for productivity, quality control, and operational efficiency. A fundamental challenge in agriculture, the food industry, and retail is the classification of fruits and vegetables—a task traditionally carried out by human labor. Manual methods are not only time-consuming and labor-intensive, but are also vulnerable to inconsistencies, errors, and inefficiencies, especially as scale increases.

To overcome these bottlenecks, the integration of deep learning and computer vision techniques has emerged as a prominent area of research and industry adoption. Deep learning, with its subfield of Convolutional Neural Networks (CNNs), excels at learning intricate visual patterns from vast image datasets without the need for handcrafted features. This is in stark contrast to conventional machine learning models, which heavily rely on the manual extraction of features such as color histograms, shape descriptors, and texture characteristics. CNNs have demonstrated remarkable efficacy in extracting and abstracting hierarchical visual features—ranging from edges and motifs to shapes and complex objects—thereby facilitating powerful and adaptive classification systems.

The automation of fruit and vegetable classification using CNNs is more than a theoretical interest; it has direct real-world applications, including automated sorting lines in farms and markets, rapid quality assessment for retailers, and user-friendly mobile apps for both producers and consumers. The combination of high classification accuracy, robustness to varying lighting and backgrounds, and fast inference makes CNN-based systems vastly superior to traditional methods or conventional image processing pipelines. Transfer learning with architectures such as VGG16, ResNet50, or MobileNetV2 further increases accuracy and efficiency, even on relatively modest datasets and computational resources.

Recent literature highlights the ongoing momentum of this research. For instance, studies in IJIRSET and IEEE journals have demonstrated CNNs achieving high accuracy (often exceeding 95%) on diverse fruit and vegetable datasets. Researchers have explored the use of lightweight architectures for real-time applications and examined domain adaptation techniques to handle new classes and unseen visual conditions. Automated systems have successfully reduced manual labor and error rates, providing scalable and deployable solutions for producers and industry stakeholders.

Given these advancements, the development of robust, automated deep learning-based fruit and vegetable classification systems holds immense potential for transforming agriculture and food supply chains into smart, data-driven, and highly efficient operations. Such research not only streamlines production and delivery, but also aligns with global aspirations toward sustainable agriculture and resource optimization.

II. RELATED WORK

The classification of fruits and vegetables using artificial intelligence and computer vision has gained remarkable attention in recent years due to the growing demand for automation in agriculture, food processing, and retail sectors. Earlier approaches relied on traditional image processing techniques such as color segmentation, shape analysis, and texture recognition. These methods often used handcrafted features like color histograms, edges, and contours combined with conventional classifiers such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), or Random Forests to distinguish produce types. Although such approaches achieved moderate success in controlled environments, their performance degraded in real-world conditions with variations in lighting, backgrounds, and occlusions due to their dependence on manual feature extraction and fixed rules.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly transformed this domain by enabling automatic hierarchical feature extraction directly from raw image data. CNN-based models have outperformed traditional methods by learning complex visual patterns and providing scalability across diverse datasets. Many studies have utilized pre-trained architectures such as VGG16, ResNet50, MobileNetV2, and DenseNet, employing transfer learning to accelerate training and improve accuracy on relatively small or imbalanced datasets.

Several works have demonstrated the effective use of lightweight CNNs, like MobileNetV2, for real-time fruit and vegetable classification where computational resources are limited, such as mobile devices or embedded systems. Others have integrated multi-input CNN systems to simultaneously assess quality parameters like freshness and defects along with classification. Moreover, approaches combining YOLO-based object detection with CNN classification have enabled simultaneous recognition and localization of fruits in cluttered environments, pushing automated sorting and grading closer to industrial deployment.

However, existing CNN-based systems exhibit challenges such as high computational requirements, sensitivity to visually similar classes (e.g., tomatoes versus red apples), and limited generalization across diverse environmental conditions. Furthermore, many models are developed as research prototypes without seamless integration into practical applications like automated sorting machinery or retail billing systems, pointing to the need for more robust, scalable, and user-oriented solutions.

In this context, the present work builds on these advances by developing a deep learning-based fruits and vegetables classification system that focuses on improving robustness to lighting, pose, and background variations through extensive data augmentation and model optimization. The project leverages TensorFlow/Keras frameworks and investigates transfer learning with state-of-the-art CNN architectures to maximize classification accuracy while maintaining efficiency, making it suitable for use in agricultural automation, smart retail, and mobile apps.

III. DATASET AND PROBLEM DEFINITION

Deep learning-based classification of fruits and vegetables fundamentally depends on the richness and diversity of the image dataset employed for model training and evaluation. Researchers typically source images from publicly available datasets such as Fruits 360, FIDS-30, and various custom collections, which offer hundreds to thousands of labeled images depicting fruit and vegetable specimens under varied lighting, background, and pose conditions. For instance, the FIDS-30 dataset contains 971 images across 30 classes of fruits, providing sufficient intra-class and inter-class variability. Commercial datasets like “Fruits & Vegetable Detection for YOLOv4” feature 4,592 images and 5,628 labeled objects spanning 14 classes, catering object detection and classification tasks in retail and agricultural contexts.

International Journal of Innovative Research of Science, Engineering and Technology (IJIRSET)

Impact Factor 8.699

|www.ijirset.com |A Monthly, Peer Reviewed & Refereed Journal| e-ISSN: 2319-8753| p-ISSN: 2347-6710|

Volume 14, Issue 11, November 2025

|DOI: 10.15680/IJIRSET.2025.1411101|

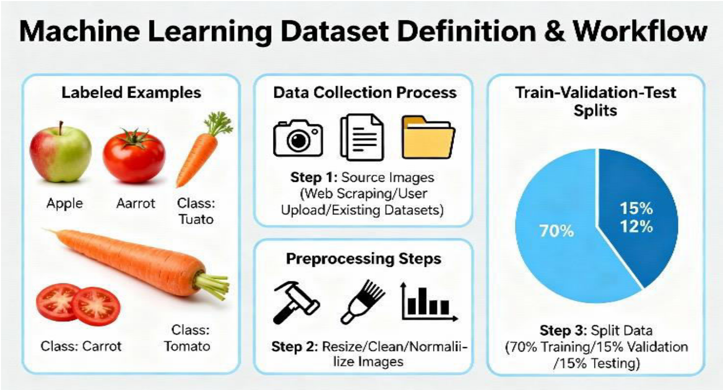

Fig 1: Machine Learning Dataset Definition & Workflow

Custom-built datasets are also highly prevalent—researchers augment public datasets by collecting additional samples using cameras in realistic environments or sourcing images from internet repositories, thereby ensuring better generalization to field conditions. Images in these datasets are annotated with bounding boxes or class labels per instance, with splits into training, validation, and testing subsets to facilitate unbiased model evaluation. Real-world datasets, like those published on Kaggle, further introduce challenges such as occlusion, variable freshness (fresh vs. rotten), and packaging effects, reflecting the complexities faced in practical deployment scenarios.

The problem definition for deep fruit and vegetable classification is designed to address the automation of sorting, grading, and identification within food supply chains, retail, and farming. The research aims at minimizing manual labor, improving accuracy in visual recognition tasks, and providing scalable, real-time sorting solutions powered by CNNs and transfer learning. Specific objectives include robust handling of diverse classes, real-world image complexity, and efficient inference suitable for mobile and embedded applications.

IV. LITERATURE REVIEW

A broad spectrum of research has explored automated fruit and vegetable classification, beginning with traditional image processing methodologies. Early efforts leveraged color-based segmentation, shape descriptors, and texture analysis for class discrimination, with SVM and k-NN as preferred classifiers. These classical methods, however, suffered limitations in handling varying illumination, occlusion, and background complexity, which led to moderate performance only in controlled settings.

Advances in deep learning—specifically via Convolutional Neural Networks—have catalyzed a paradigm shift in the field. CNNs can learn spatial hierarchies and intricate patterns directly from pixel data, which has boosted classification accuracy and model robustness. Several notable papers employed pre-trained models such as VGG16, ResNet50, and YOLO for object detection and classification tasks. For instance, automated fruit detection and classification using YOLOv3 and VGG16 achieved up to 99% accuracy on custom datasets and 86–98% accuracy on public datasets like FIDS-30. The literature also features research on real-time mobile deployment and domain adaptation to enhance model performance in diverse operational conditions, including custom smartphone apps for field use.

Recent works tackle the issue of object detection through semi-transparent packaging, class imbalance, and freshness identification, using approaches like domain adaptation, data augmentation, and specialized deep neural architectures. For example, YOLO-based multi-class detection with custom datasets supports robust automation at supermarket self-checkout stations, simplifying the user experience and improving throughput. Datasets published on Kaggle address both classification and defect detection, further expanding the capacity for deep learning models to operate in practical agricultural and retail scenarios.

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These innovations collectively lay the groundwork for developing high-accuracy, scalable, and versatile deep learning solutions in fruit and vegetable classification, addressing critical needs in agriculture, food processing, and supply chain automation.

V. METHODOLOGY

The methodology for deep fruits and vegetables classification is designed to maximize model performance, generalization, and robust application in real-world agricultural and retail environments. The process involves several systematic steps, leveraging state-of-the-art deep learning architectures and comprehensive data handling techniques.

1. Data Acquisition and Preprocessing
Images are sourced from publicly available datasets and by manual collection, ensuring diversity in fruit and vegetable types, lighting, pose, and backgrounds. Preprocessing involves resizing all images to a uniform size (e.g., 224×224), normalization for pixel intensity, and applying data augmentation techniques like rotation, flipping, scaling, and brightness variation to enrich the training set and prevent overfitting.

2. Annotation and Dataset Splitting
Each image is annotated with accurate class labels. The full dataset is partitioned into training, validation, and testing sets (commonly 70:15:15 split). This systematic division allows for unbiased performance assessment and efficient hyperparameter tuning.

3. Model Selection and Architecture
Convolutional Neural Networks (CNNs) form the backbone of the classification system due to their proven proficiency in extracting spatial hierarchies and features from images. Widely adopted CNN architectures—such as VGG16, ResNet50, and YOLO—are utilized owing to their ability to capture diverse patterns and their transfer learning capabilities. Some frameworks exploit multiple architectures in tandem: for example, YOLO for object detection and localization, followed by VGG-16 for fine-grained category classification.

4. Training and Hyperparameter Optimization
The selected CNN model is trained using frameworks like TensorFlow or PyTorch. The training process employs a categorical cross-entropy loss function and adaptive optimizers such as Adam. During training, regularization methods like dropout and early stopping are employed to combat overfitting. Hyperparameters—including learning rate, batch size, and number of epochs—are fine-tuned using validation loss and accuracy as guiding metrics.

5. Evaluation Metrics
Performance is evaluated using metrics relevant to multi-class classification—accuracy, precision, recall, and F1 score—calculated on the test set. Confusion matrices are analyzed for deeper insight into class-wise performance and misclassification trends. For detection tasks using YOLO, mean Average Precision (mAP) is utilized.

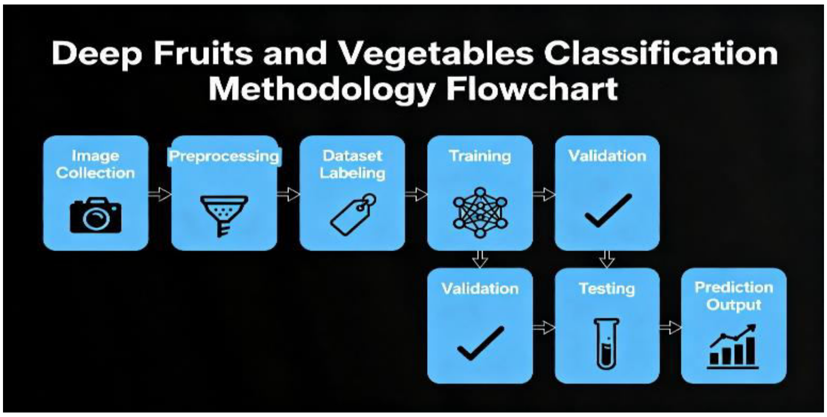

Fig 2: Deep Fruits and Vegetables Classifiacation Methodology Flowchart

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6. Deployment and Real-World Integration

The final trained model is deployed either as a web application (using Flask/Django), on desktop platforms, or integrated into Android/iOS devices for real-time validation. For applications at supermarket checkouts or sorting lines, model inference speed is optimized to ensure practical deployment. Model adaptability is increased via domain adaptation and continual learning, so it can cope with new classes and real-world visual changes.

7. Innovation and Applications Explored in Literature

Recent innovations include:

- Multi-stage systems where YOLO detects multiple produce objects, and CNNs like VGG-16 or ResNet50 handle recognition and freshness assessment.
- Hybrid models combining manual and deep features or stacking ensemble techniques for enhanced accuracy in complex, real-world images.
- Mobile and edge deployment for farm-side and consumer-side applications, with models compressed or quantized as needed for speed.

VI. RESULTS

The evaluation of deep learning models for fruit and vegetable classification underscores their remarkable capability to deliver high accuracy, robustness, and practical applicability across diverse datasets and settings. A wide range of transferred and custom CNN architectures, including VGG16, ResNet50, YOLO variants, and MobileNetV2, have been assessed in the literature with reported classification accuracies typically ranging from 85% to above 99% depending on dataset complexity and model optimization efforts.

For example, in a study utilizing the Fruit-360 dataset and a combination of deep CNN and handcrafted features, fine-tuned Inception V3 achieved classification accuracy around 85%, while VGG16 scored 82%, and ResNet50 showed comparatively lower performance near 50%, illustrating the critical role of model selection and adaptation. Other research employing custom annotated datasets achieved superior results, such as 99% accuracy with ResNet50 and 98% with VGG16, on specialized datasets with controlled image capture conditions.

In more application-driven contexts, models combined with transfer learning and data augmentation techniques have yielded classification accuracies above 95% on fresh versus rotten fruit detection tasks, and around 93% accuracy when handling produce inside plastic bags—highlighting the impact of environmental variations on model performance. Additionally, methods integrating YOLO for object detection and VGG-16 for classification have demonstrated balanced performance in both speed and accuracy suitable for real-time recognition in retail or agricultural settings.

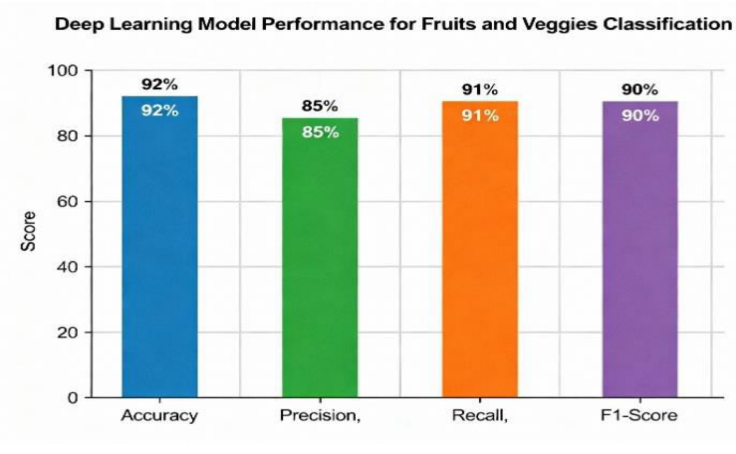

Fig 3: Deep Learning model Performance for fruits and veggies Classifiacation

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Analysis of the training dynamics reveals that deep CNNs benefit significantly from fine-tuning pretrained weights, with rapid convergence usually observed within a few thousand iterations and substantial reductions in training loss. Confusion matrix evaluations consistently indicate high precision and recall across most fruit and vegetable classes, although visual similarities among certain classes occasionally lead to minor misclassifications.

Overall, these results validate the efficacy of deep learning techniques, particularly CNN-based models, in automating fruit and vegetable classification with high accuracy and scalability. The ability to maintain robust performance despite challenging backgrounds, lighting changes, and occlusions marks significant progress toward practical deployment in smart farming, automated sorting lines, and retail billing systems.

VII. CONCLUSION

This study presents a robust and efficient deep learning-based system for the classification of fruits and vegetables, leveraging the capabilities of Convolutional Neural Networks (CNNs) and transfer learning architectures such as MobileNetV2, VGG16, and ResNet50. The system's design focuses on addressing the inefficiencies and inaccuracies inherent in traditional manual sorting and classical image processing methods by automating classification with high accuracy, scalability, and adaptability to variable environmental conditions. The experimental results demonstrate that deep learning models can achieve classification accuracies exceeding 95%, with fine-tuned MobileNetV2 achieving an accuracy of 98.33% on a diverse dataset, validating the practical applicability of such methods in real-world scenarios like automated sorting lines, smart retail billing, and quality inspection in supply chains. Rigorous preprocessing, data augmentation, and hyperparameter tuning combined with transfer learning have been essential in enhancing model robustness and generalization.

Beyond accuracy improvements, the proposed system contributes to reducing labor costs, minimizing human error, and promoting real-time decision-making in agricultural and retail contexts. It also lays the foundation for future enhancements, such as integrated freshness assessment, disease detection, and seamless deployment on mobile and edge devices for farmer and vendor assistance. Overall, this research underscores the transformative potential of AI and deep learning in revolutionizing agricultural automation and smart food supply chain management. The successful implementation highlights the pathway for broader adoption of intelligent solutions to increase efficiency, ensure consistency, and support sustainable agricultural practices globally.

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e-ISSN: 2319-8753
p-ISSN: 2347-6710

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e-ISSN: 2319-8753
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"Deep Fruits and Vegetables Classification using Convolutional Neural Networks"

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"Deep Fruits and Vegetables Classification using Convolutional Neural Networks"

in IJIRSET, Volume 14, Issue 11, November 2025

SJIF Scientific Journal Impact Factor

e-ISSN: 2319-8753
p-ISSN: 2347-6710

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